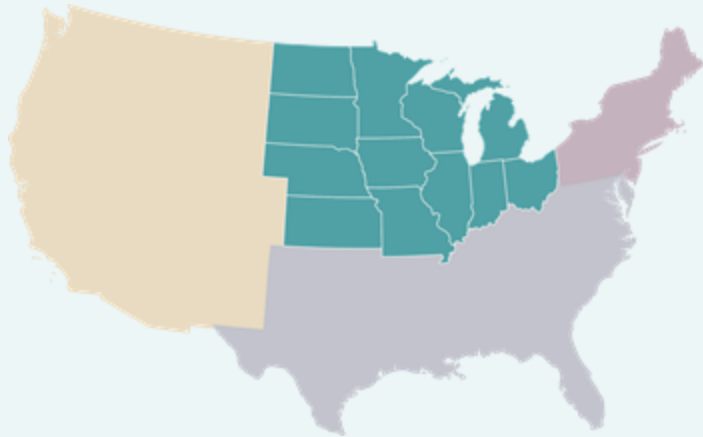


# Spatial Data Science for Land Conservation in the United States



**Christina Dennis** [cldennis@wisc.edu](mailto:cldennis@wisc.edu)

Advised by **Holly Gibbs**

- [illegible]

### Thesis text word cloud

# Students!

- **Raines:**
  - Grassland conversion at the property level
- **Sam:**
  - Mapping US organic farms
- **Lucas:**
  - Mapping US Farm payments



Raines Lucas



Sam Kuemmel



Lucas Palomino



# LAND CONSERVATION IS CLIMATE ACTION

**NATURE KNOWS BEST**  
when we work for nature, nature works for us.



## LET THE TREES GROW

Trees are carbon-absorbing superheroes, and Wisconsin has billions of them. Healthy, well-managed forests can support the forest products industry and store carbon better than any technology.



## CONNECT WILDLIFE CORRIDORS

Too many vital wildlife habitats are isolated from one another. Islands of conservation. Land trusts protect habitat corridors that support wildlife migrations and allow plant and animal populations to thrive as the climate changes.



## FARM FOR TOMORROW

Regenerative agriculture restores the land and provides food for the dinner table. Wisconsin has more than 15 million acres of farmland. Rotational grazing, cover crops, and less frequent tilling all help to make farming a climate solution.



## CARE FOR WETLANDS

Wetlands protect our communities from floods. And they act as a filter and a sponge, cleaning water as it passes through and soaking up pollution. Land trusts work to protect intact wetlands and restore those that have been harmed.



## KEEP IT WILD

Our wild places can be a powerful defense against the effects of climate change. Undisturbed land absorbs more carbon from the atmosphere, supports a wider array of plants and wildlife, and offers natural protection from threats like flooding, erosion, and pollution.



## RESTORE GRASSLANDS

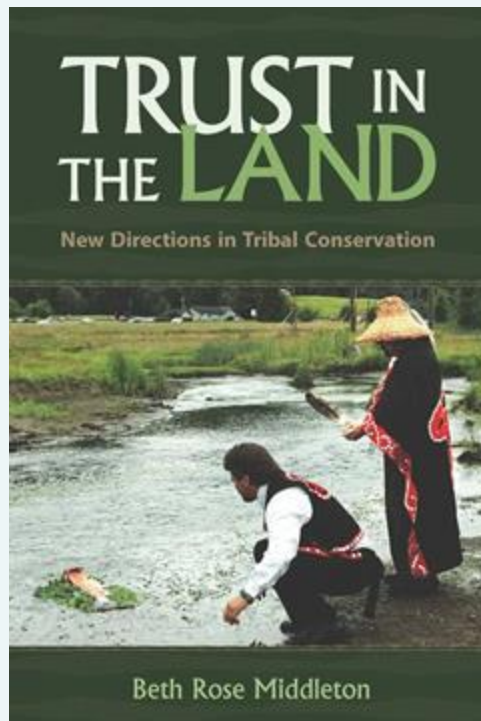
90% of our native grasses have been lost. Land trusts work to protect what's left and to restore grasslands where we can. The deep roots of prairie grasses store tremendous amounts of carbon and support impressive populations of wildlife.



GATHERINGWATERS.ORG



@GATHERINGWATERS



# Land Conservation

- **Habitat loss** threatens **biodiversity** and ecosystem **function**
- Natural areas provide clean water, carbon storage, and pollination
- **Conversion** continues: 1M+ acres/year of grassland lost in the Great Plains
- Most remaining natural areas exist as **fragmented** patches across **working** landscapes

LETTER

## Carbon emissions from cropland expansion in the United States

Seth A Spawn<sup>1,2</sup>, Tyler J Lark<sup>2</sup> and Holly K Gibbs<sup>1,2</sup>

<sup>1</sup> Department of Geography, University of Wisconsin—Madison, United States of America

<sup>2</sup> Center for Sustainability and the Global Environment, University of Wisconsin—Madison, United States of America



Land Use Policy

Volume 97, September 2020, 104727



## Protecting our prairies: Research and policy actions for conserving America's grasslands

Tyler J. Lark<sup>a,b</sup>

RESEARCH ARTICLE | SUSTAINABILITY SCIENCE



## Tropical forests were the primary sources of new agricultural land in the 1980s and 1990s

H. K. Gibbs<sup>a</sup>, A. S. Ruesch<sup>a</sup>, F. Achard<sup>a</sup>, and J. A. Foley<sup>a</sup> [Authors Info & Affiliations](#)

Edited by B. L. Turner, Arizona State University, Tempe, AZ, and approved July 30, 2010 (received for review September 22, 2009)

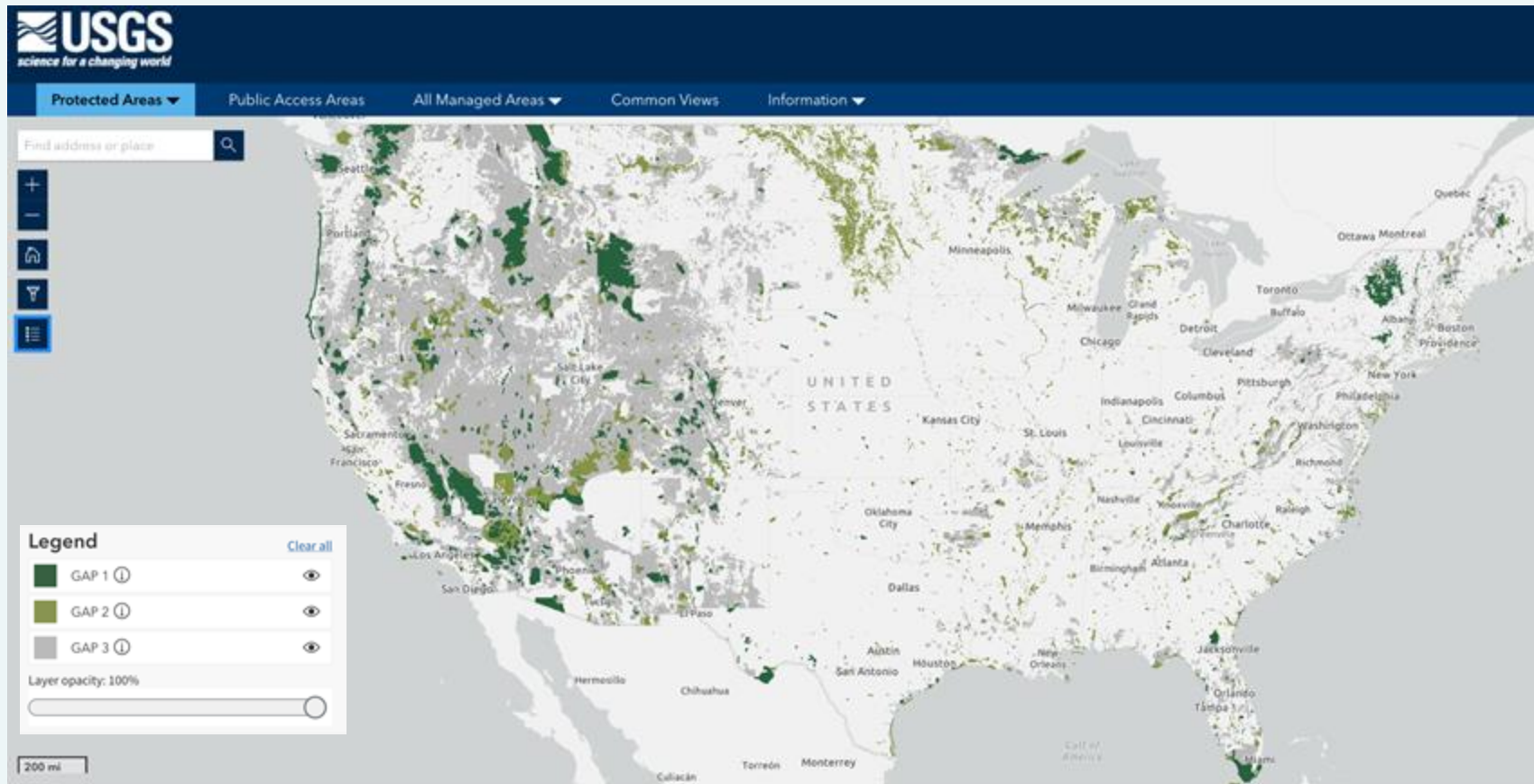
August 31, 2010 | 107 (38) 16732–16737 | <https://doi.org/10.1073/pnas.0910275107>

## Nature's Kidneys: the Role of Wetland Reserve Easements in Restoring Water Quality

Nicole Karwowski and Marin Skidmore \*



# Protected Areas in the United States



# Private Land Dominance

- ~**60%** of US land (1.4 billion acres) is **privately** owned
- Private landowners manage:
  - 90% of U.S. cropland
  - 70% of forest-use land
  - 66% of grassland/pasture



# Land Conservation: **trade-offs**

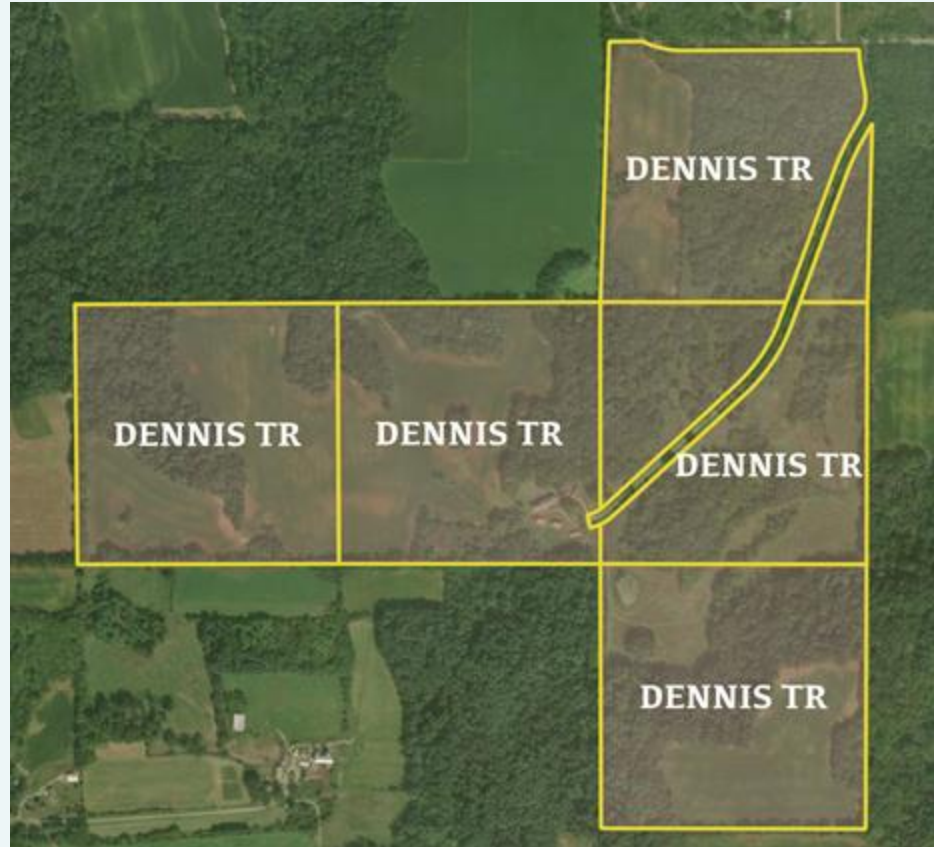
- **System bias:** Policy and markets **favor production** over habitat.
- Core **trade-off:** Food and income vs. biodiversity and ecosystem health.





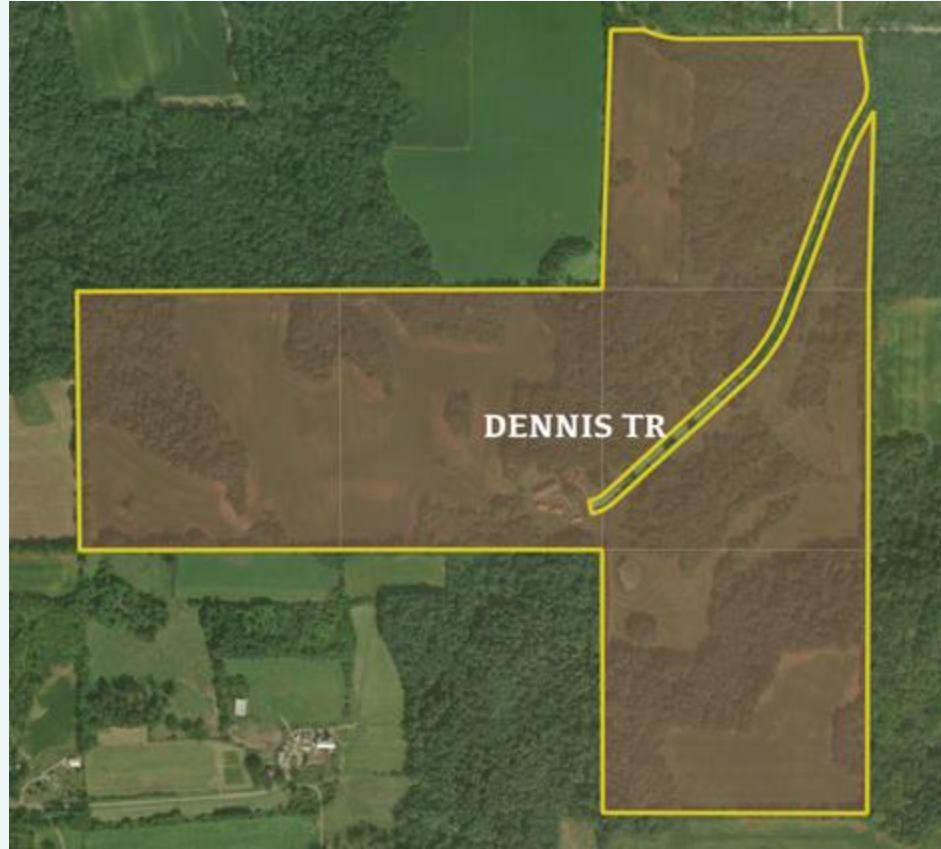
# Parcels are **individual** records of land **ownership**

**Five** Parcels,  
All owned  
by the  
Dennis Trust



Together, all parcels owned by the same landowner  
become a **Property**

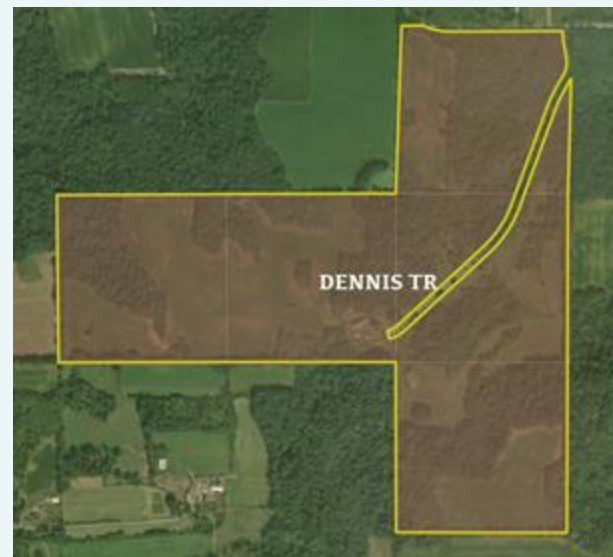
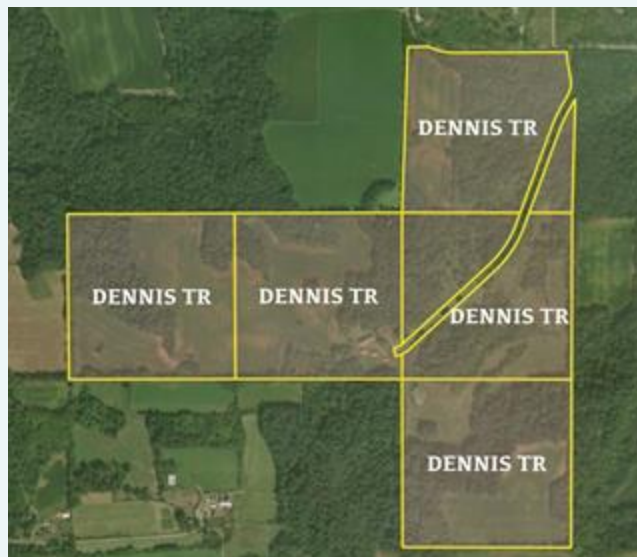
Five Parcels,  
**One** Property  
Owned  
by the  
Dennis Trust



# Chapter 1: From parcels to properties

Matching attributes and boundaries of tax records across the United States

*In Review at  
Nature  
Scientific Data!!!!*

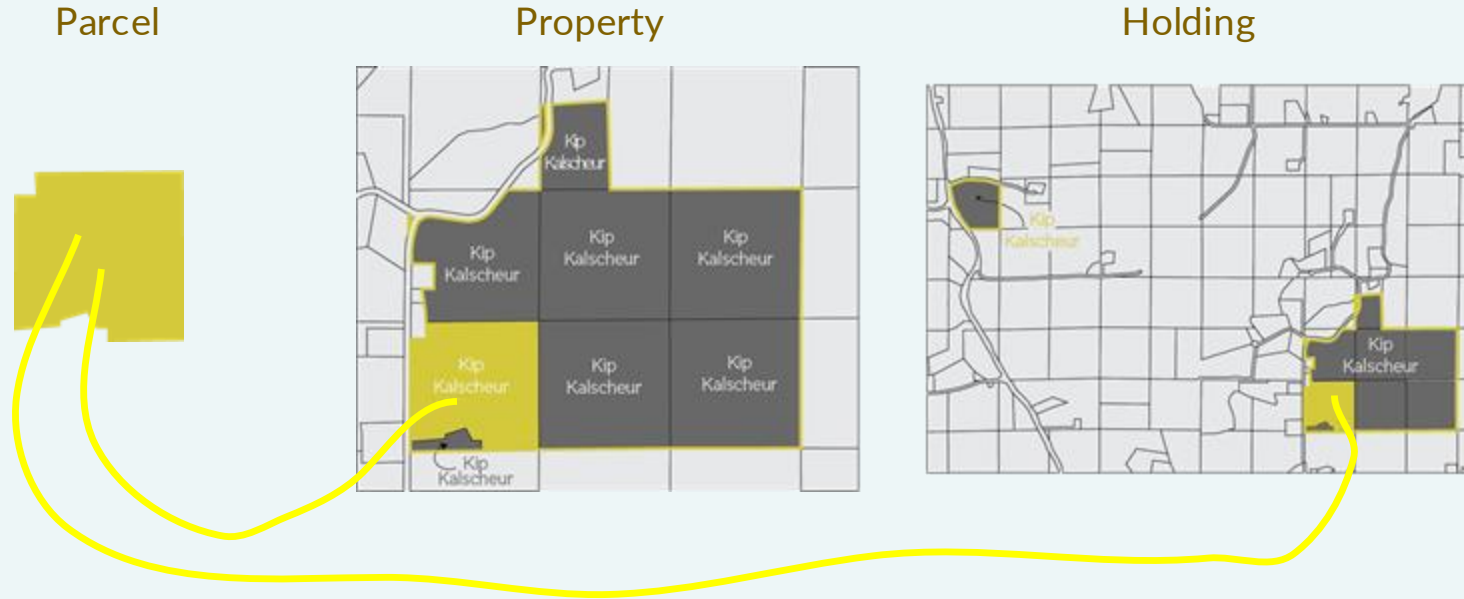


# Using owner **name** and **mailing address** To create **entities**





# Parcels & Properties & Holdings (locations may be different)

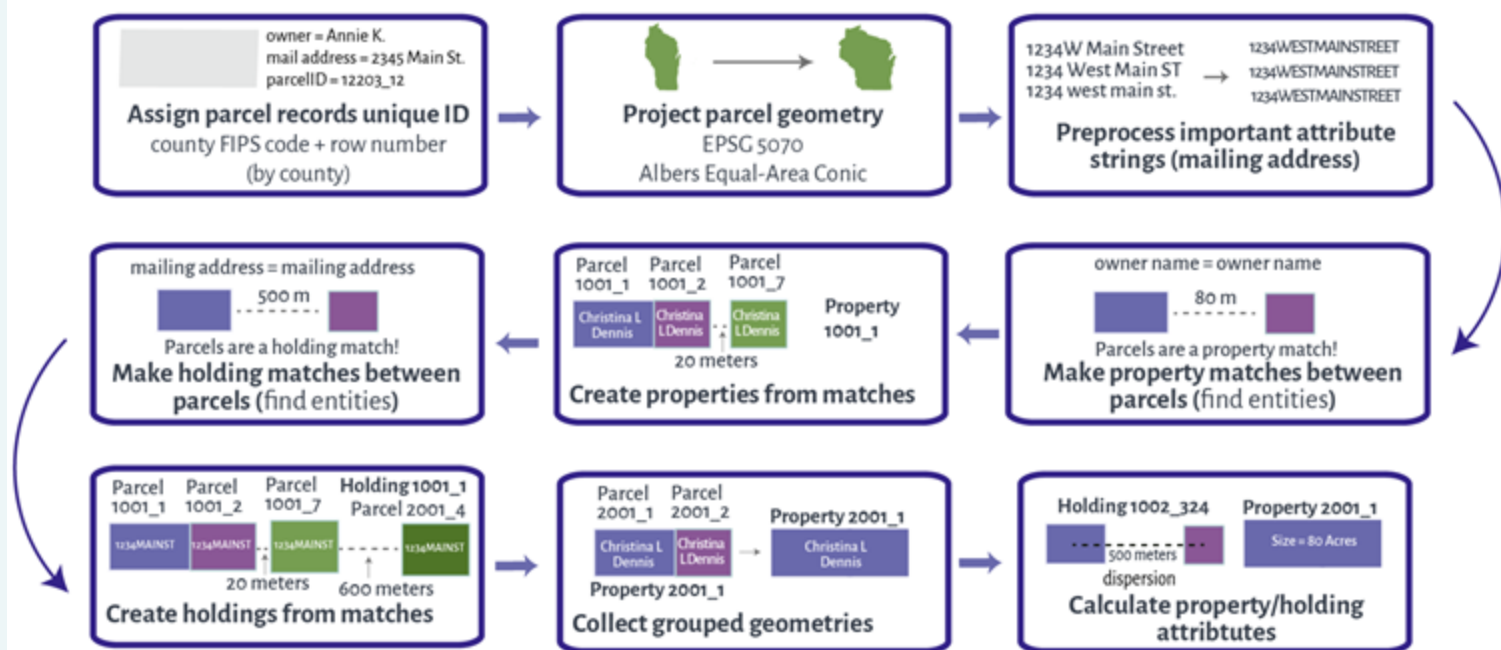


# Parcels to Properties

Parcel Source – [Regrid.com](https://www.regrid.com)

- 150 million+ parcels in the US
- Entity Matching ← Data science

## Data Pipeline



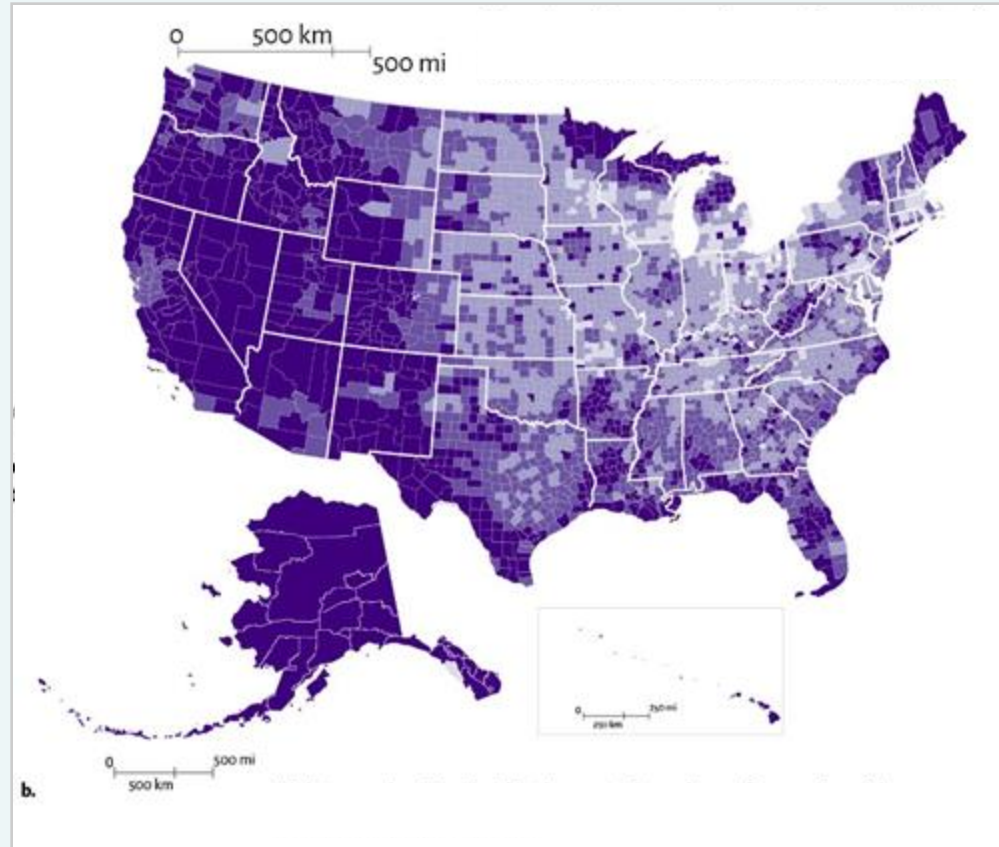
# Results from paper: summaries of #s

(a.)

| Region | #of Parcels | #of Properties | Avg Area | Median Area | Avg (AW) Localness | Avg (AW) Urban | % Land Local > 0.5 | % Land Local > 0.0 |
|--------|-------------|----------------|----------|-------------|--------------------|----------------|--------------------|--------------------|
| W      | 1           | 24,129,731     | 12.37    | 0.2         | 0.1                | 0.04           | 9.86               | 9.86               |
| W      | 2-5         | 1,899,201      | 70.59    | 1.43        | 0.25               | 0.02           | 25.34              | 29.18              |
| W      | 6+          | 185,256        | 3895.29  | 23.03       | 0.09               | 0              | 6.78               | 31.3               |
| MW     | 1           | 23,538,654     | 6.11     | 0.29        | 0.41               | 0.09           | 40.95              | 40.95              |
| MW     | 2-5         | 3,767,539      | 49.66    | 2.87        | 0.46               | 0.02           | 46.55              | 49.38              |
| MW     | 6+          | 381,372        | 418.15   | 123.31      | 0.38               | 0.03           | 38.36              | 49.46              |
| S      | 1           | 42,846,733     | 4.56     | 0.32        | 0.36               | 0.14           | 36.34              | 36.34              |
| S      | 2-5         | 5,411,213      | 35.25    | 2.6         | 0.4                | 0.05           | 39.52              | 42.16              |
| S      | 6+          | 435,833        | 488.53   | 20.03       | 0.2                | 0.04           | 19.62              | 28.61              |
| NE     | 1           | 18,615,164     | 3.34     | 0.27        | 0.44               | 0.23           | 43.88              | 43.88              |
| NE     | 2-5         | 1,579,050      | 22.4     | 1.13        | 0.44               | 0.09           | 43.68              | 47.8               |
| NE     | 6+          | 97,859         | 305.62   | 11.72       | 0.21               | 0.1            | 20.56              | 36.88              |

# Example: Are property owners **local** or **absentee**?

- **Light** Purple = Most Property owners are LOCAL
- **Dark** Purple = Most Property owners are ABSENTEE

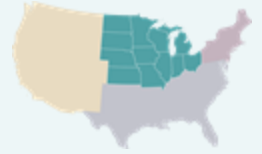




Why such **rectangle**?

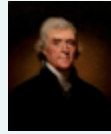


# Private property rights are strong with this one



John Locke 1632-1704, UK

The right to **land** and resources gained through one's labor — were a **natural right** of all men.



Thomas Jefferson 1743-1826, VA

## Public Land Survey System

- Document land taken
- Sell land & **raise money** for war debt
- Collect taxes \$\$\$\$



Figure 1: Land holdings of colonies in 1781. (VirginiaPlaces.org)

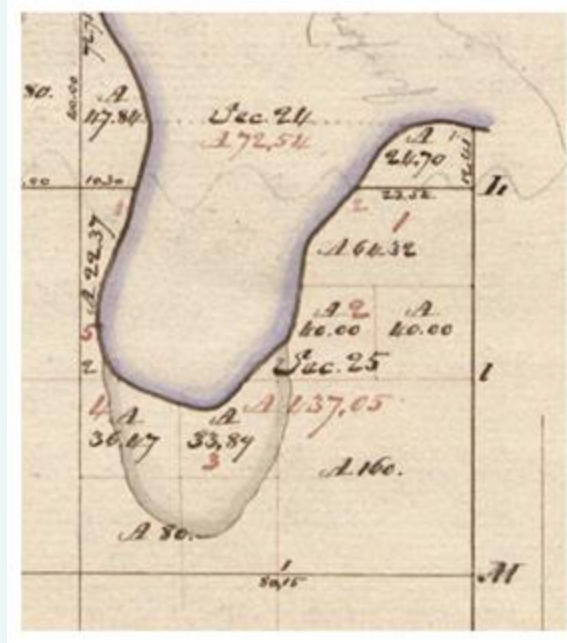
## Land Surveyor



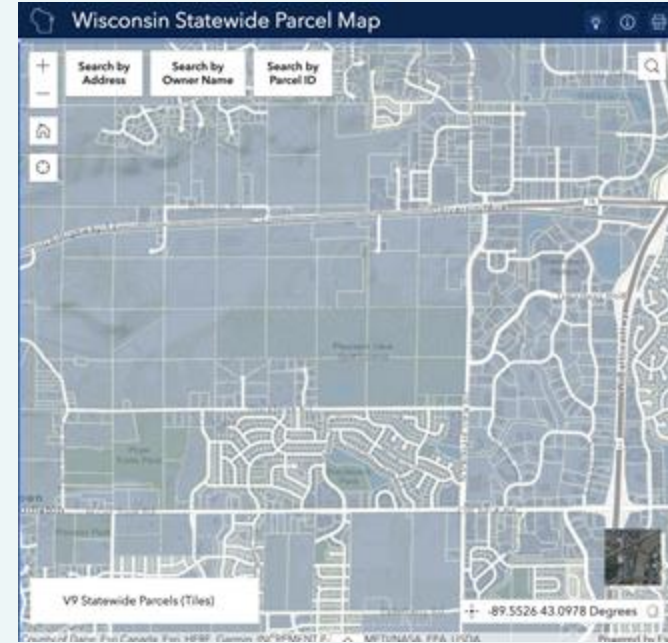
# PLSS Grid → Parcel Data

|                        |                        |                       |                       |
|------------------------|------------------------|-----------------------|-----------------------|
| NW 1/4<br>of<br>NW 1/4 | NE 1/4<br>of<br>NW 1/4 | NE 1/4                |                       |
| SW 1/4<br>of<br>NW 1/4 | SE 1/4<br>of<br>NW 1/4 |                       |                       |
| N 1/2 of SW 1/4        |                        | W 1/2<br>of<br>SE 1/4 | E 1/2<br>of<br>SE 1/4 |
| S 1/2 of SW 1/4        |                        |                       |                       |

Public Land Survey System grid layout

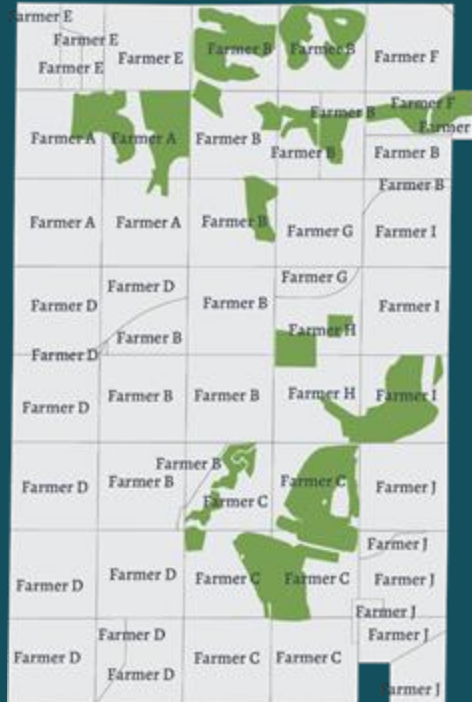


1800's surveyor map



Present-day parcel data (Wisconsin, State cartographer's office)

## From individual parcels:



Conserved  
Land



Property  
with CRP

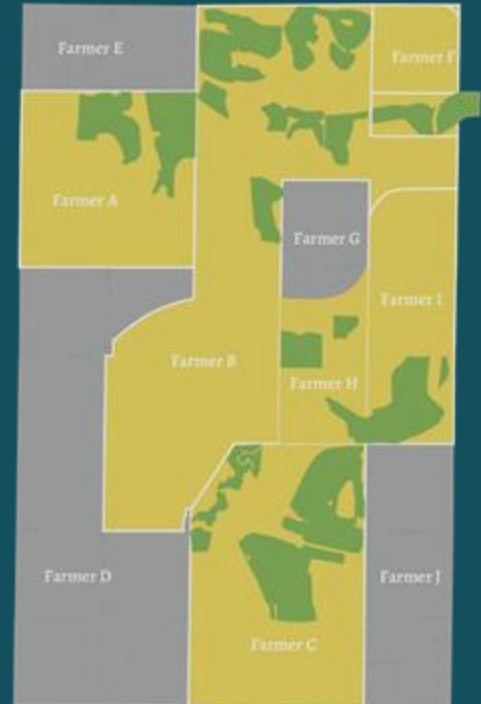


Property  
without CRP



Original  
Parcels

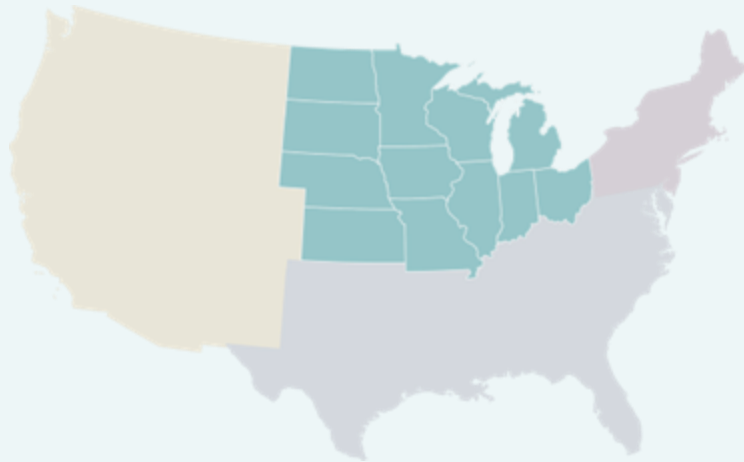
## To cohesive properties:



# Chapter 2: Property-Level Conservation

## Habitat Protection on Private Lands in the Rural Midwest

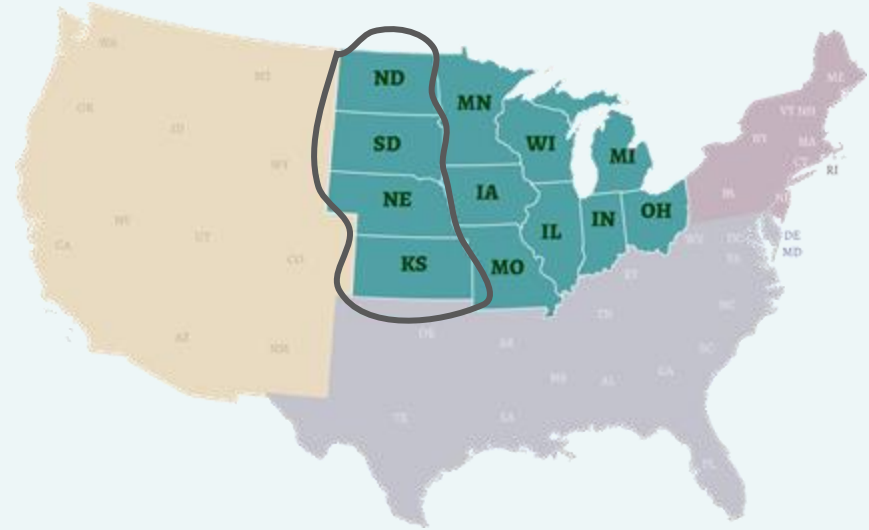
- What **property traits** are associated with the **presence** and **extent** of conserved **habitat** on rural private properties in the **Midwest**?





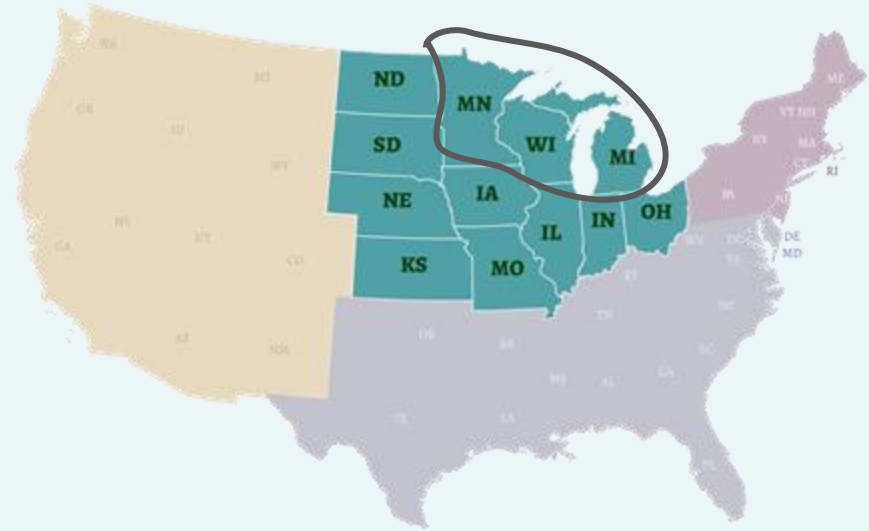
# Why the **Midwest**?

- **Diverse** agroecological systems:
  - Northern Plains
    - 51% cropland, 41% pasture



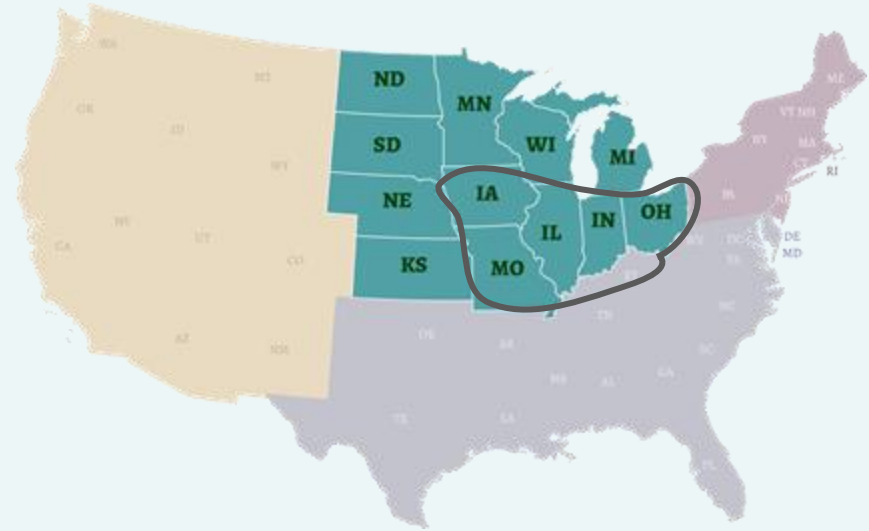
# Why the **Midwest**?

- **Diverse** agroecological systems:
  - Northern Plains
    - 51% cropland, 41% pasture
  - Lake States
    - 42% forest-use land



# Why the **Midwest**?

- **Diverse** agroecological systems:
  - Northern Plains
    - 51% cropland, 41% pasture
  - Lake States
    - 42% forest-use land
  - Corn Belt
    - 54% cropland



# Why the **Midwest**?

- **Diverse** agroecological systems:
  - Northern Plains
    - 51% cropland, 41% pasture
  - Lake States
    - 42% forest-use land
  - Corn Belt
    - 54% cropland
- Overwhelmingly **privately** owned (<5% federal lands)
- Ideal setting to examine conservation across **ownership** types and **land uses**



# Conservation Pathway # 1: **Formal**

- Formally **enrolled** land
  - **Voluntary** programs
    - balancing private property **rights**  
& **conservation** goals



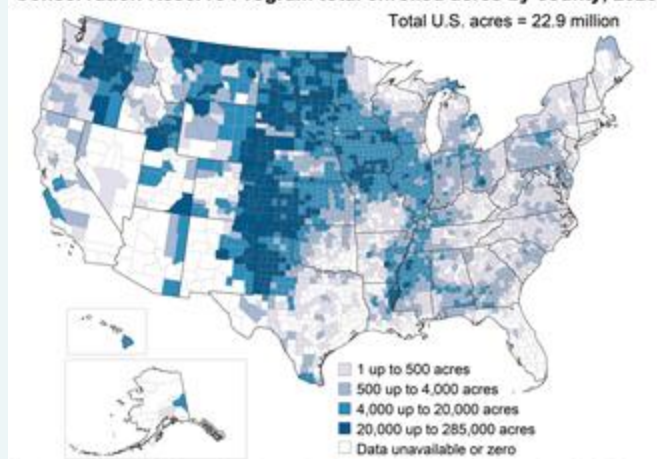
# Conservation Pathway # 1: **Formal**

- Formally **enrolled** land

# The Conservation Reserve Program: USDA Working Lands Conservation

- CRP
- Working lands program
  - **Convert** sensitive land **out** of **cropland** and into **vegetative** cover
  - 10-15 year **contracts**
    - Annual rental **payments**

Conservation Reserve Program total enrolled acres by county, 2023



Note: Total acres include continuing and newly enrolled acres as of September 30, 2023.  
Source: USDA, Economic Research Service using data from USDA, Farm Service Agency.



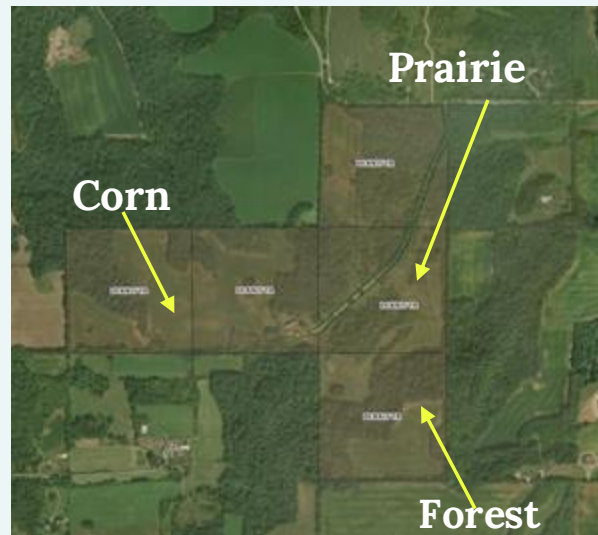
# Conservation Easements

- Formally **enrolled** land
  - Conservation **Easements**
    - Permanent legal restrictions
    - One-time compensation or tax benefits
    - Landowners work with groups like the Nature Conservancy, NRCS



# Conservation Pathway #2: **Informal**

- Informal (conserved?) habitat
  - **Stewardship** decisions, intentional or incidental
  - Remnant vegetation, field edges, shelterbelts, riparian buffers, grasslands, grazing land, wetlands, woodlots
  - Also referred to as:
    - Semi-natural
    - Land-sparing
    - Marginal
    - Unused
    - Non-crop



# Research Gap

- Existing research focuses on **individual** programs or small **case** studies
- Unit of analysis:
  - **Isolated** parcel data **obscures** decision making
  - **Census** tracts, **county** level are coarse
  - **Pixel** data lacks socioeconomic context
- With property data >> **full portfolio** of
  - cultivated land, formally conserved, informally conserved

# Sample: Rural, Private Midwest

- First regional analysis integrating formal + informal conservation
  - Excludes urban areas
  - Excludes Gap 1-3 Protected Areas besides conservation easements
- 7.5 million **rural private** properties
  - 414 million acres in the **Midwest**



# Data Sources

- Outcome Unit: **Property boundaries** I created from Regrid parcel data (2024)
  - **CRP** enrollment **polygons**: USDA FSA (2023) \*via Tyler Lark!
  - Conservation **easements polygons**: USGS PAD-US 4.0
  - Land **cover**: USDA Cropland Data Layer (10m resolution)
- 
- **Informal** habitat = natural habitat not in a formally enrolled program (spatially explicit subtraction)

# What is **correlated** with **Informal Habitat**?

## **Economic, Biophysical, Geographic, Ownership** predictors

- 17 property **characteristics**
  1. **Economic**: Trade-offs, incentives, pressures
  2. **Land** Characteristics: Biophysical constraints
  3. **Geography**: Landscape context
  4. **Ownership**: Structure

# Variables (1/4): Economics

| Variable            | Description                                                                               |
|---------------------|-------------------------------------------------------------------------------------------|
| Land Value          | Estimated fair market <b>value</b> of the land                                            |
| Subsidies           | Does the property receive any disaster relief or commodity <b>subsidies</b> from the FSA? |
| CRP Rental Rate     | Maximum <b>rate</b> receive if property has CRP otherwise county average                  |
| Formal Conservation | Is there any <b>formally</b> conserved land on the property (CRP or CEs)                  |
| Organic Farm        | Is there an <b>organic</b> farm on the property                                           |



## Variables (2/4): Land

| Variable                  | Description                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------|
| Corn/Soy Share            | Amount of property that is <b>corn</b> or <b>soy</b>                                    |
| Crop Diversity            | The number of <b>distinct crops (not just row crops)</b> cultivated within the property |
| Mean Slope                | The mean <b>slope</b> of the property                                                   |
| Irrigated Share           | Amount of property that is <b>irrigated</b>                                             |
| Soil Productivity (NCCPI) | Inherent capacity of soils to produce dryland commodity crops                           |



# Variables (3/4): Geographic

| Variable                                | Description                                                                                                |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------|
| Distance to closest formal practice     | Distance from property centroid to closest formal conservation practice centroid not on the same property. |
| Distance to closest urban area or place | Distance to closest census-designated urban area/place.                                                    |



# Variables (4/4): Ownership

| Variable                          | Description                                                                                           |
|-----------------------------------|-------------------------------------------------------------------------------------------------------|
| Property Size                     | <b>Size</b> of the property                                                                           |
| Holding Dispersion                | <b>Longest distance</b> between vertices of all parcels within holding property belongs to            |
| Localness (0=absentee, 1 = local) | Average <b>localness</b> of the owner mailing addresses within the property (weighted by parcel area) |
| Owner is an Organization          | Likely owner <b>type</b> is an <b>organization</b> (LLC, Inc, Farm)                                   |





# Methods: Summaries & Regressions

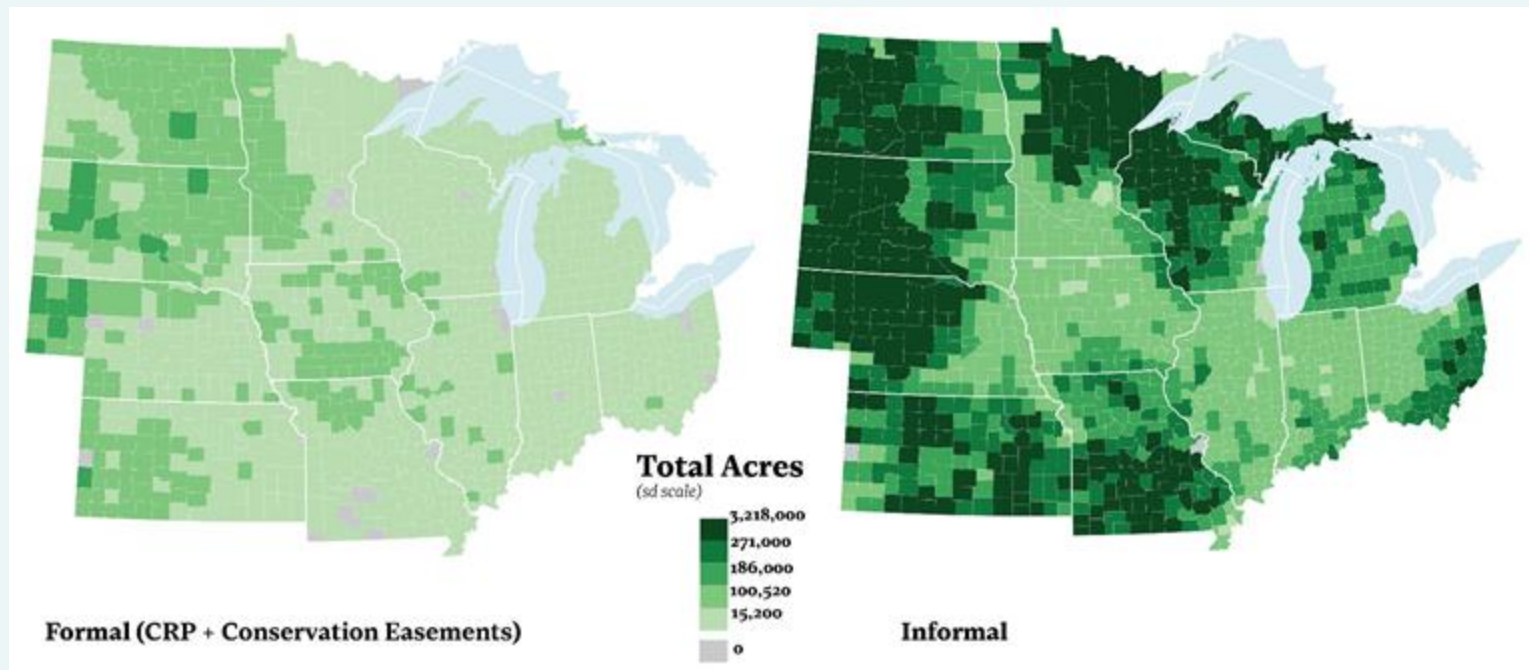
- Normalized differences (compare properties)
- **Presence** models (binary logit)
- **Extent** models (OLS, beta regression)
  - For the **Midwest**
  - *Heterogeneity Analyses:*
    - By **State**
    - By **forest, grassland, wetland**

# Scale of Informal Habitat

- **Informal Habitat:** 195.5M acres (48% of all rural MW)
  - 4.9M properties **contain** at least 1 acre of informal habitat
  - **Grassland:** 113M acres (58% of informal)
  - **Forest:** 59.6M acres (31%)
  - **Wetland:** 19.5M acres (10%)
- **Formal Conservation:** 13.6M acres (3.3%)
  - 4% of properties **enrolled** in formal programs
    - **CRP:** 10.7M acres
    - **Easements:** 3.1M acres

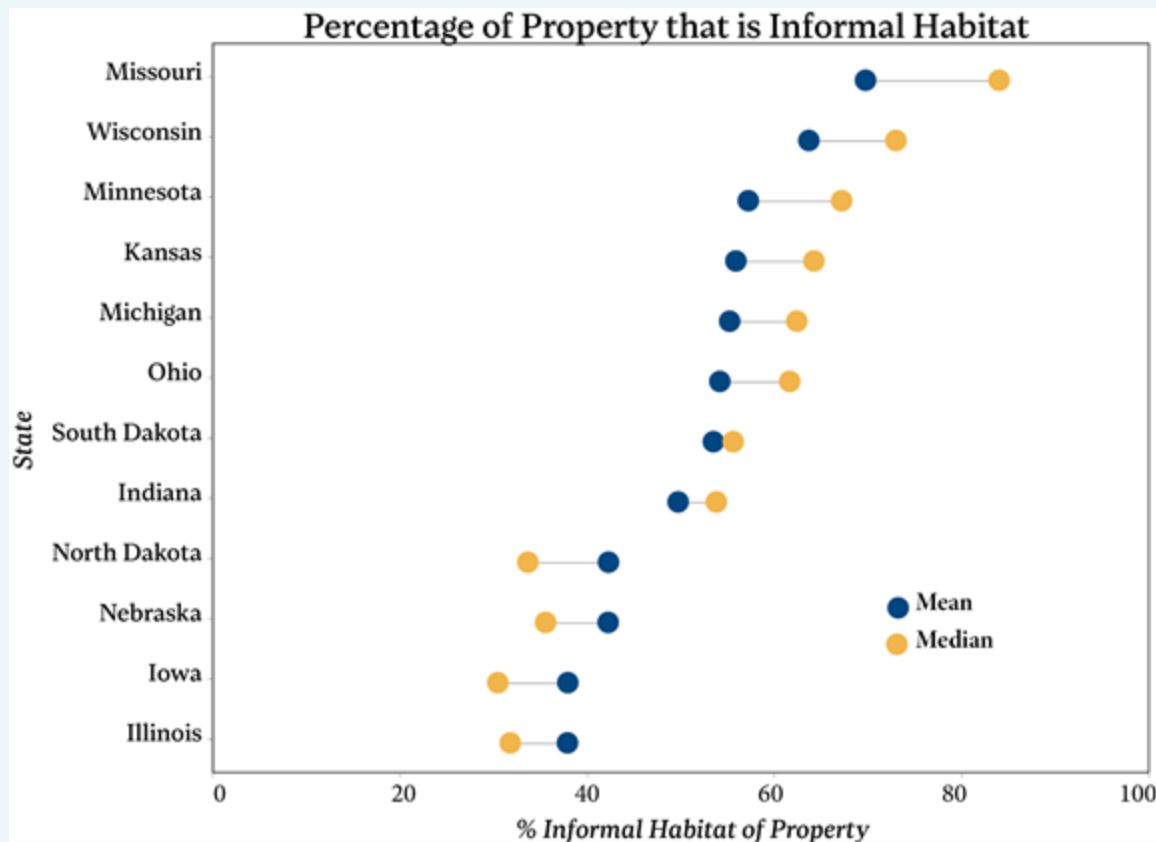
# Habitat by Acreage

- **Informal Habitat:** 195.5M acres (48% of all rural MW)
- **Formal Conservation:** 13.6M acres (3.3%)



# Private Rural MW Properties are on **average** 66% informal habitat

- Variation by State



- I did a lot
- I will **highlight** the results I'm discussing
- Will be focusing in **informal** habitat, not formal (see paper if interested)

# Properties **with** and **without** at least **1 acre** of informal habitat

## Comparing **variables** with *Normalized Differences*

|                                            |
|--------------------------------------------|
| Land Value (ln(\$/ha))                     |
| Crop Diversity (# of crops)                |
| Distance to urban area (km)                |
| CRP rental rate (\$/acre/yr)               |
| Has formal conservation                    |
| Mean slope                                 |
| Soil Productivity                          |
| Property area (ac)                         |
| Subsidies Received (\$/prop)               |
| Owner is Organization                      |
| Absenteeism                                |
| Has organic farm                           |
| Irrigated Area (%)                         |
| Holding area (ac)                          |
| Distance to closest enrolled practice (km) |
| Holding dispersion (km)                    |
| Share of Corn/Soy cover                    |



**Positive** normalized diff = attribute is **HIGHER** in properties **with** informal habitat

| Variable                                   | No Informal           |             | Has Informal          |             | Norm. Diff |
|--------------------------------------------|-----------------------|-------------|-----------------------|-------------|------------|
|                                            | Mean<br>(N=2,472,115) | SD          | Mean<br>(N=4,949,561) | SD          |            |
| Land Value (ln(\$/ha))                     | 9.950                 | 1.004       | 9.363                 | 0.886       | -0.621**   |
| Crop Diversity (# of crops)                | 0.465                 | 0.886       | 1.144                 | 1.764       | +0.486**   |
| Distance to urban area (km)                | 3.627                 | 3.755       | 4.620                 | 4.202       | +0.249**   |
| CRP rental rate (\$/acre/yr)               | 152.895               | 80.854      | 134.335               | 78.430      | -0.233**   |
| Has formal conservation                    | 0.013                 | 0.114       | 0.054                 | 0.226       | +0.229**   |
| Mean slope                                 | 2.842                 | 2.945       | 3.521                 | 3.164       | +0.222**   |
| Soil Productivity                          | 575.501               | 191.306     | 535.195               | 177.982     | -0.218**   |
| Property area (ac)                         | 13.624                | 46.646      | 75.695                | 434.562     | +0.201**   |
| Subsidies Received (\$/prop)               | 330.401               | 5995.340    | 778.159               | 7730.782    | +0.065*    |
| Owner is Organization                      | 0.056                 | 0.230       | 0.071                 | 0.257       | +0.062*    |
| Absenteeism                                | 0.573                 | 0.494       | 0.599                 | 0.487       | +0.053*    |
| Has organic farm                           | 0.000                 | 0.013       | 0.001                 | 0.037       | +0.044     |
| Irrigated Area (%)                         | 0.026                 | 0.147       | 0.030                 | 0.144       | +0.026     |
| Holding area (ac)                          | 106035.417            | 1422441.474 | 88078.723             | 1290649.195 | -0.013     |
| Distance to closest enrolled practice (km) | 380.352               | 421.083     | 375.206               | 413.292     | -0.012     |
| Holding dispersion (km)                    | 52.942                | 233.891     | 53.643                | 219.210     | +0.003     |
| Share of Corn/Soy cover                    | 0.156                 | 0.565       | 0.155                 | 0.296       | -0.001     |

Normalized diff. =  $(\bar{X}_{with} - \bar{X}_{without}) / \sqrt{(s_{with}^2 + s_{without}^2) / 2}$ .

\*\*|Norm. Diff.| > 0.1; \*|Norm. Diff.| > 0.05.

# Variables that are **substantially** different (diff > 0.1)

| Variable                                   | No Informal           |             | Has Informal          |             | Norm.<br>Diff |
|--------------------------------------------|-----------------------|-------------|-----------------------|-------------|---------------|
|                                            | Mean<br>(N=2,472,115) | SD          | Mean<br>(N=4,949,561) | SD          |               |
| Land Value (ln(\$/ha))                     | 9.950                 | 1.004       | 9.363                 | 0.886       | -0.621**      |
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| Irrigated Area (%)                         | 0.026                 | 0.147       | 0.030                 | 0.144       | +0.026        |
| Holding area (ac)                          | 106035.417            | 1422441.474 | 88078.723             | 1290649.195 | -0.013        |
| Distance to closest enrolled practice (km) | 380.352               | 421.083     | 375.206               | 413.292     | -0.012        |
| Holding dispersion (km)                    | 52.942                | 233.891     | 53.643                | 219.210     | +0.003        |
| Share of Corn/Soy cover                    | 0.156                 | 0.565       | 0.155                 | 0.296       | -0.001        |

Normalized diff. =  $(\bar{X}_{with} - \bar{X}_{without}) / \sqrt{(s_{with}^2 + s_{without}^2)/2}$ .

\*\*|Norm. Diff.| > 0.1; \*|Norm. Diff.| > 0.05.

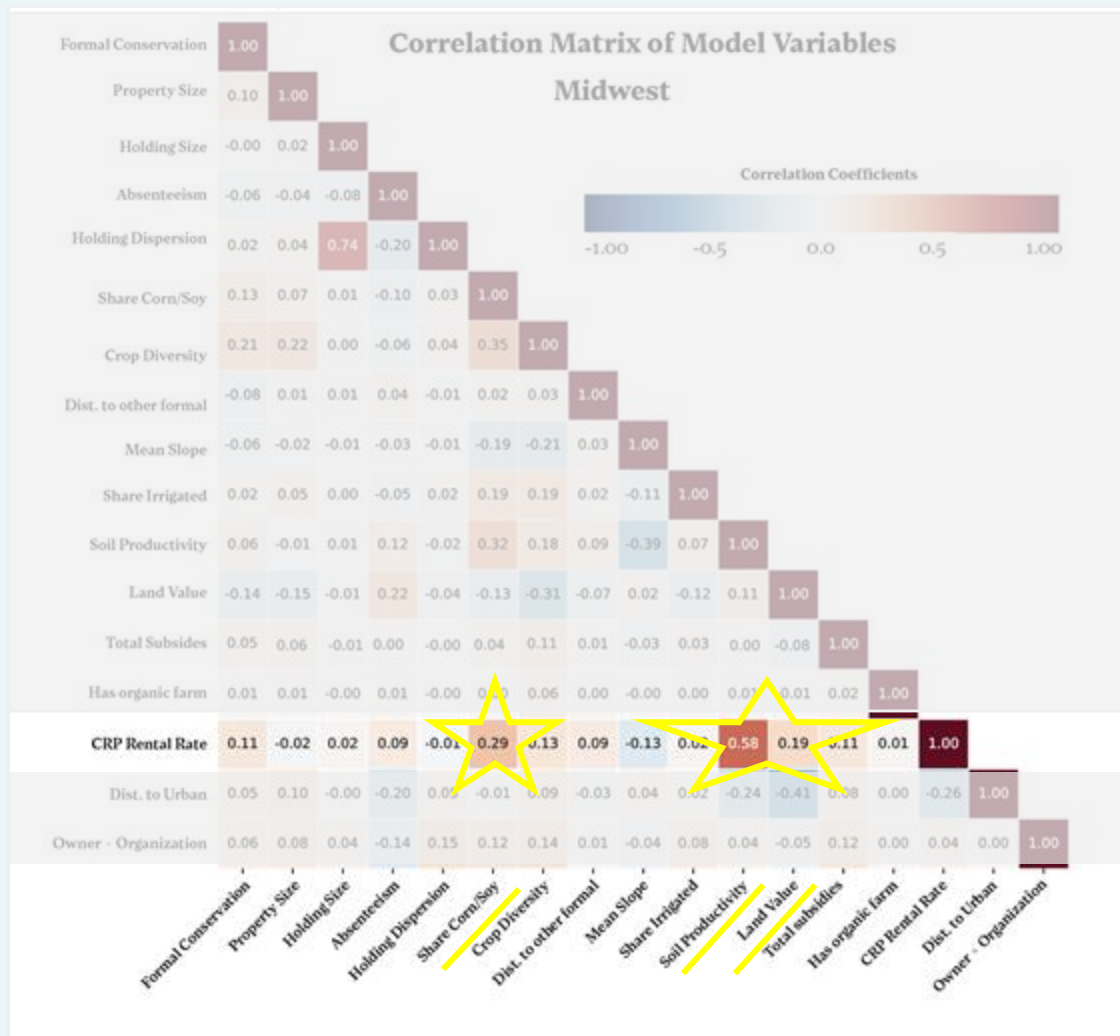
# Variables that are **substantially** different (diff > 0.1)

**Negative** =  
is *higher* in  
properties  
**without**  
informal habitat

| Variable                                   | No Informal<br>Mean<br>(N=2,472,115) | SD          | Has Informal<br>Mean<br>(N=4,949,561) | SD          | Norm.<br>Diff |
|--------------------------------------------|--------------------------------------|-------------|---------------------------------------|-------------|---------------|
| Land Value (ln(\$/ha))                     | 9.950                                | 1.004       | 9.363                                 | 0.886       | -0.621**      |
| CRP rental rate (\$/acre/vr)               | 152.895                              | 80.854      | 134.335                               | 78.430      | -0.233**      |
| Soil Productivity                          | 575.501                              | 191.306     | 535.195                               | 177.982     | -0.218**      |
| Subsidies Received (\$/prop)               | 330.401                              | 5995.340    | 778.159                               | 7730.782    | +0.065*       |
| Owner is Organization                      | 0.056                                | 0.230       | 0.071                                 | 0.257       | +0.062*       |
| Absenteeism                                | 0.573                                | 0.494       | 0.599                                 | 0.487       | +0.053*       |
| Has organic farm                           | 0.000                                | 0.013       | 0.001                                 | 0.037       | +0.044        |
| Irrigated Area (%)                         | 0.026                                | 0.147       | 0.030                                 | 0.144       | +0.026        |
| Holding area (ac)                          | 106035.417                           | 1422441.474 | 88078.723                             | 1290649.195 | -0.013        |
| Distance to closest enrolled practice (km) | 380.352                              | 421.083     | 375.206                               | 413.292     | -0.012        |
| Holding dispersion (km)                    | 52.942                               | 233.891     | 53.643                                | 219.210     | +0.003        |
| Share of Corn/Soy cover                    | 0.156                                | 0.565       | 0.155                                 | 0.296       | -0.001        |

Normalized diff. =  $(\bar{X}_{with} - \bar{X}_{without}) / \sqrt{(s_{with}^2 + s_{without}^2)/2}$ .

\*\*|Norm. Diff.| > 0.1; \*|Norm. Diff.| > 0.05.



# Variables that are **substantially** different (diff > 0.1)

**Positive** =  
is higher in  
properties  
**with**  
informal habitat

| Variable                                   | No Informal<br>Mean<br>(N=2,472,115)SD |             | Has Informal<br>Mean<br>(N=4,949,561)SD |             | Norm.<br>Diff |
|--------------------------------------------|----------------------------------------|-------------|-----------------------------------------|-------------|---------------|
| Crop Diversity (# of crops)                | 0.465                                  | 0.886       | 1.144                                   | 1.764       | +0.486**      |
| Distance to urban area (km)                | 3.627                                  | 3.755       | 4.620                                   | 4.202       | +0.249**      |
| Has formal conservation                    | 0.013                                  | 0.114       | 0.054                                   | 0.226       | +0.229**      |
| Mean slope                                 | 2.842                                  | 2.945       | 3.521                                   | 3.164       | +0.222**      |
| Property area (ac)                         | 13.624                                 | 46.646      | 75.695                                  | 434.562     | +0.201**      |
| Subsidies Received (\$/prop)               | 330.401                                | 5995.340    | 778.159                                 | 7730.782    | +0.065*       |
| Owner is Organization                      | 0.056                                  | 0.230       | 0.071                                   | 0.257       | +0.062*       |
| Absenteeism                                | 0.573                                  | 0.494       | 0.599                                   | 0.487       | +0.053*       |
| Has organic farm                           | 0.000                                  | 0.013       | 0.001                                   | 0.037       | +0.044        |
| Irrigated Area (%)                         | 0.026                                  | 0.147       | 0.030                                   | 0.144       | +0.026        |
| Holding area (ac)                          | 106035.417                             | 1422441.474 | 88078.723                               | 1290649.195 | -0.013        |
| Distance to closest enrolled practice (km) | 380.352                                | 421.083     | 375.206                                 | 413.292     | -0.012        |
| Holding dispersion (km)                    | 52.942                                 | 233.891     | 53.643                                  | 219.210     | +0.003        |
| Share of Corn/Soy cover                    | 0.156                                  | 0.565       | 0.155                                   | 0.296       | -0.001        |

Normalized diff. =  $(\bar{X}_{with} - \bar{X}_{without}) / \sqrt{(s^2_{with} + s^2_{without})/2}$ .

\*\*|Norm. Diff.| > 0.1; \*|Norm. Diff.| > 0.05.

# Methods: Summaries & **Regressions**

- Normalized differences (compare properties)
- *Regression*: Put those variables in conversation with each other
- **Presence**: Does property have conservation?
- **Extent**: How much land is conserved?
  - **Percentage** of Property

## For **regressions**, variables **standardized** for comparability

- Variables have **different units** (acres, dollars, percentages etc)
  - Hard to compare
- Variables were standardized to be comparable
  - **Continuous** ~ mean 0, SD 1
    - Interpret coefficients as **one-SD** change
  - **Binary** stays 0/1
    - Interpret as a **0 to 1 change**

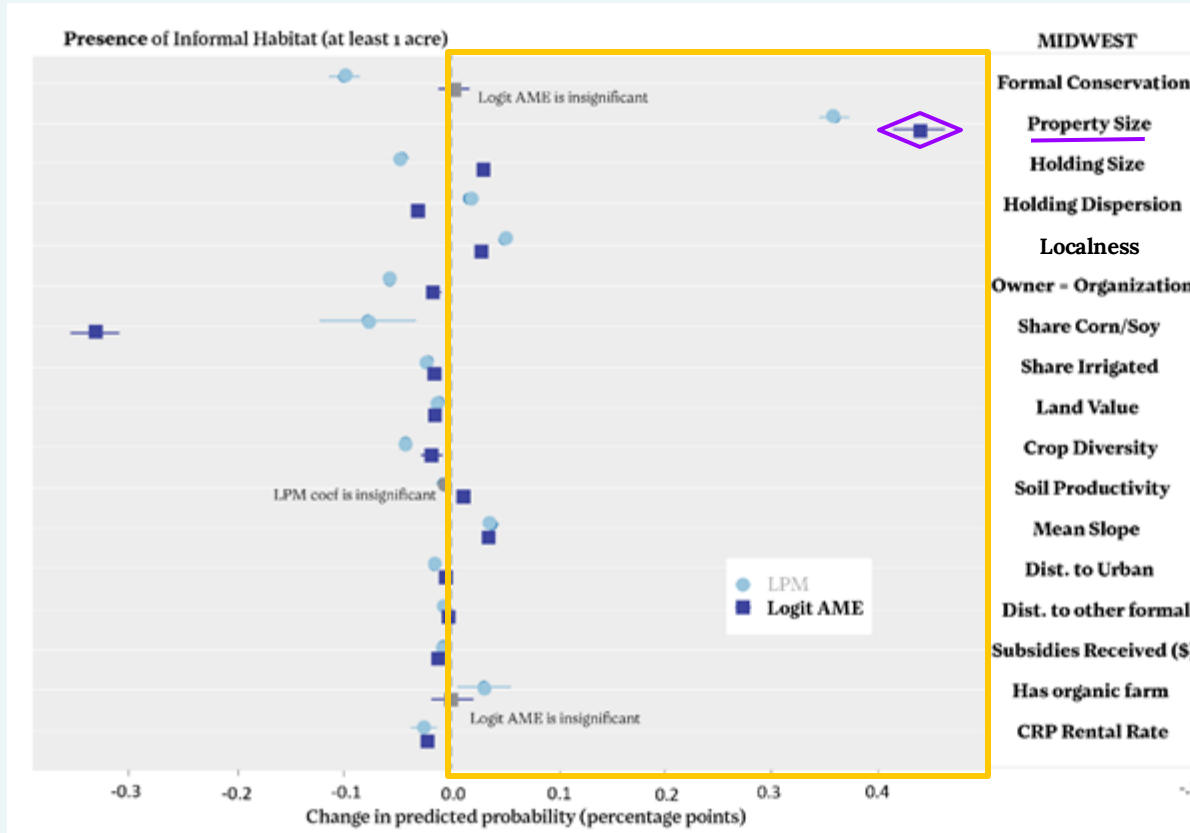
What is **correlated** with the **presence** and **extent** of at least one acre of informal habitat?



Starting with the **Presence** of at least **1 acre** of **informal habitat**

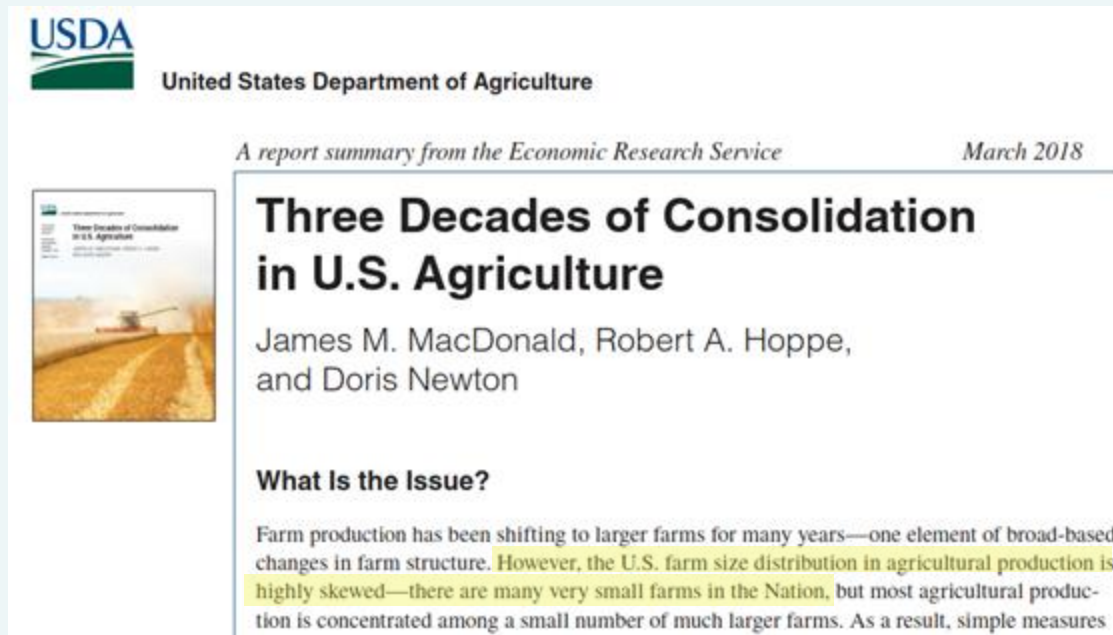
## Property Size

Positively  
predicts  
presence

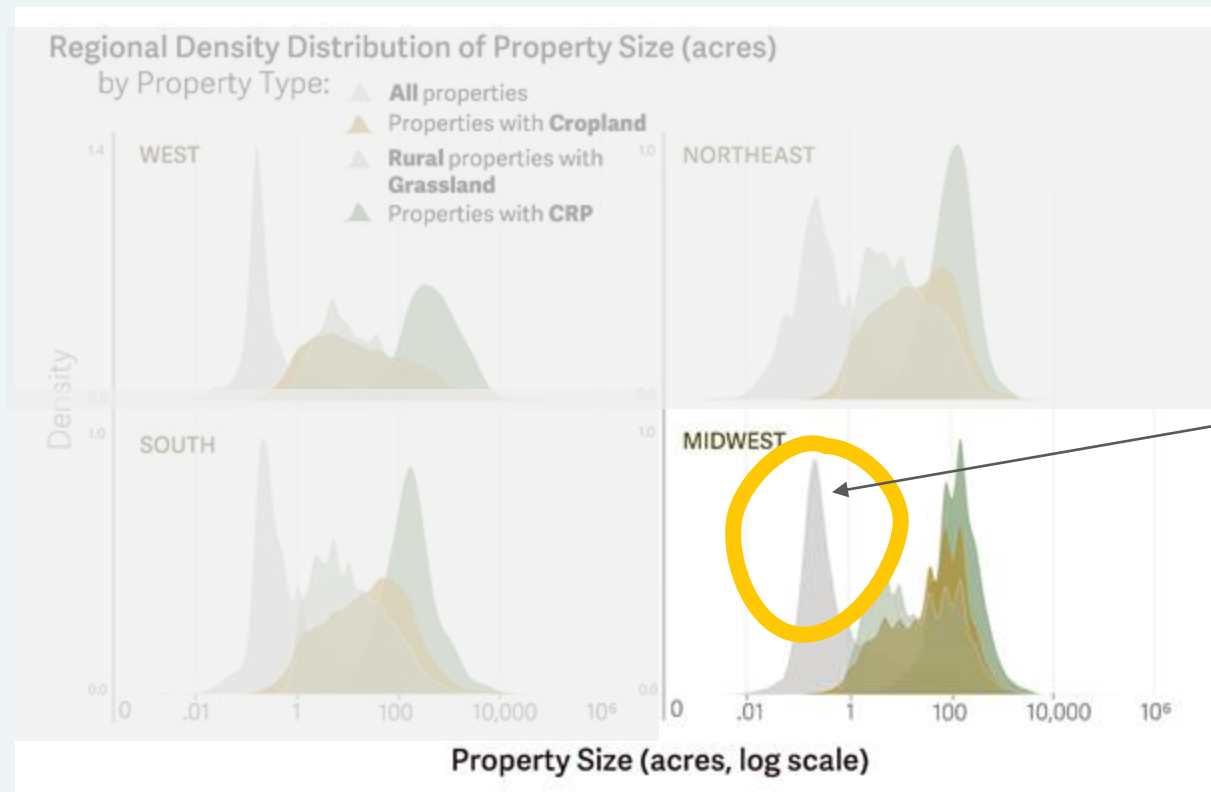


# Property size is **skewed** (lots of small ones)

- Well documented that there are many many small *farms*
- Same is true for properties



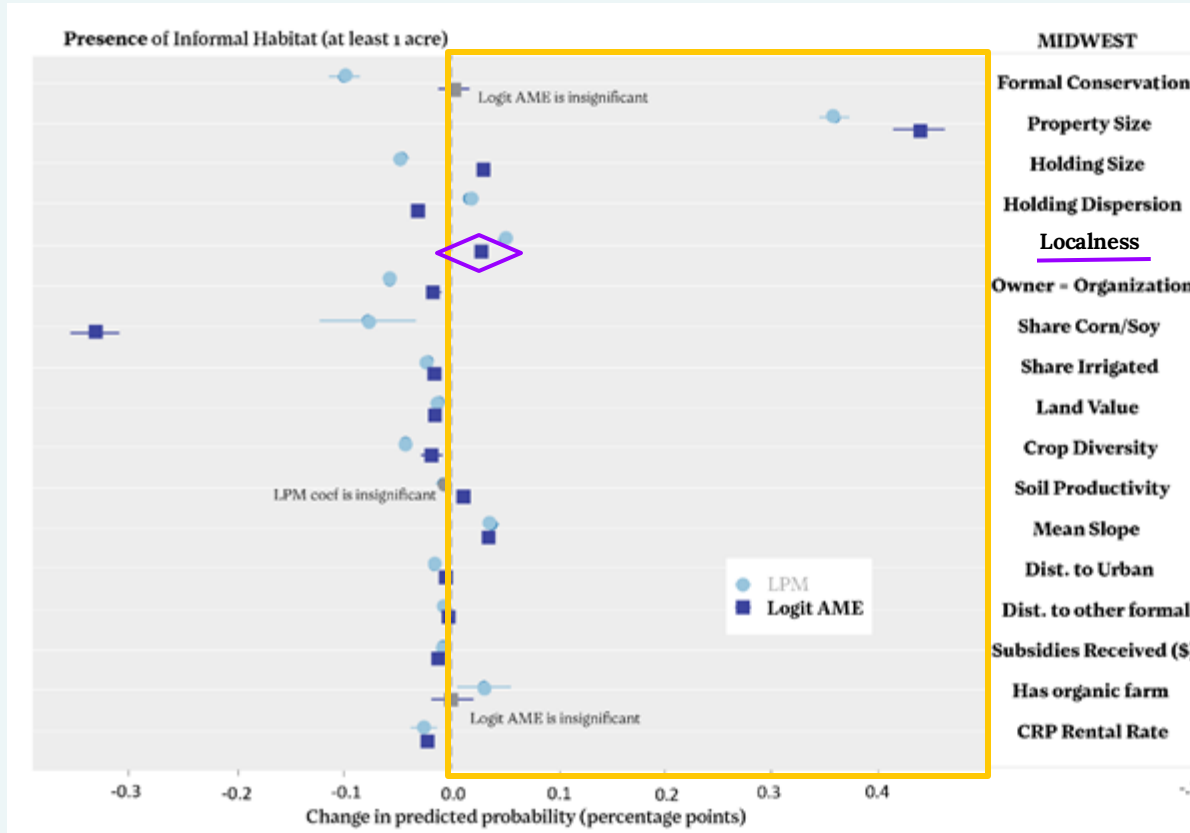
Property size is skewed (lots of small ones)  
so I log transform it in the models (ln)



Small properties  
(urban,  
residential)

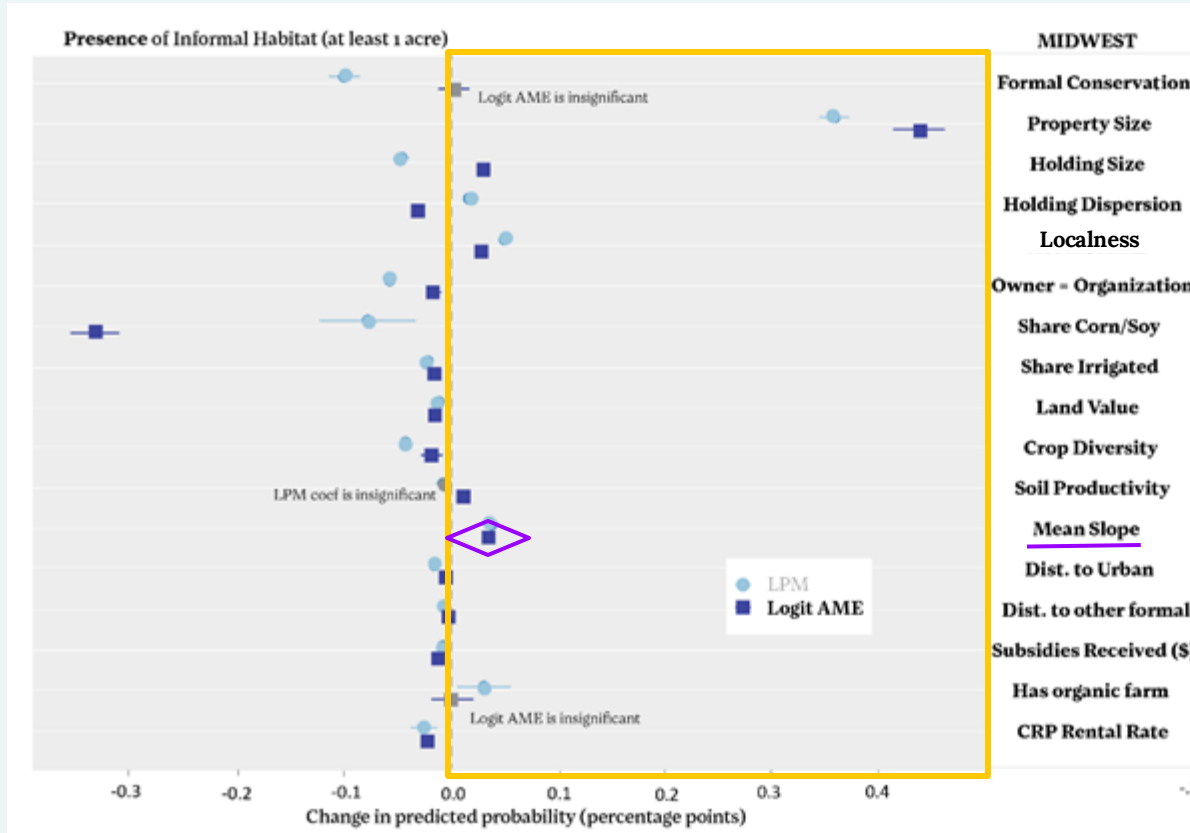
Localness  
0=absentee, 1= local

Positively  
predicts  
presence



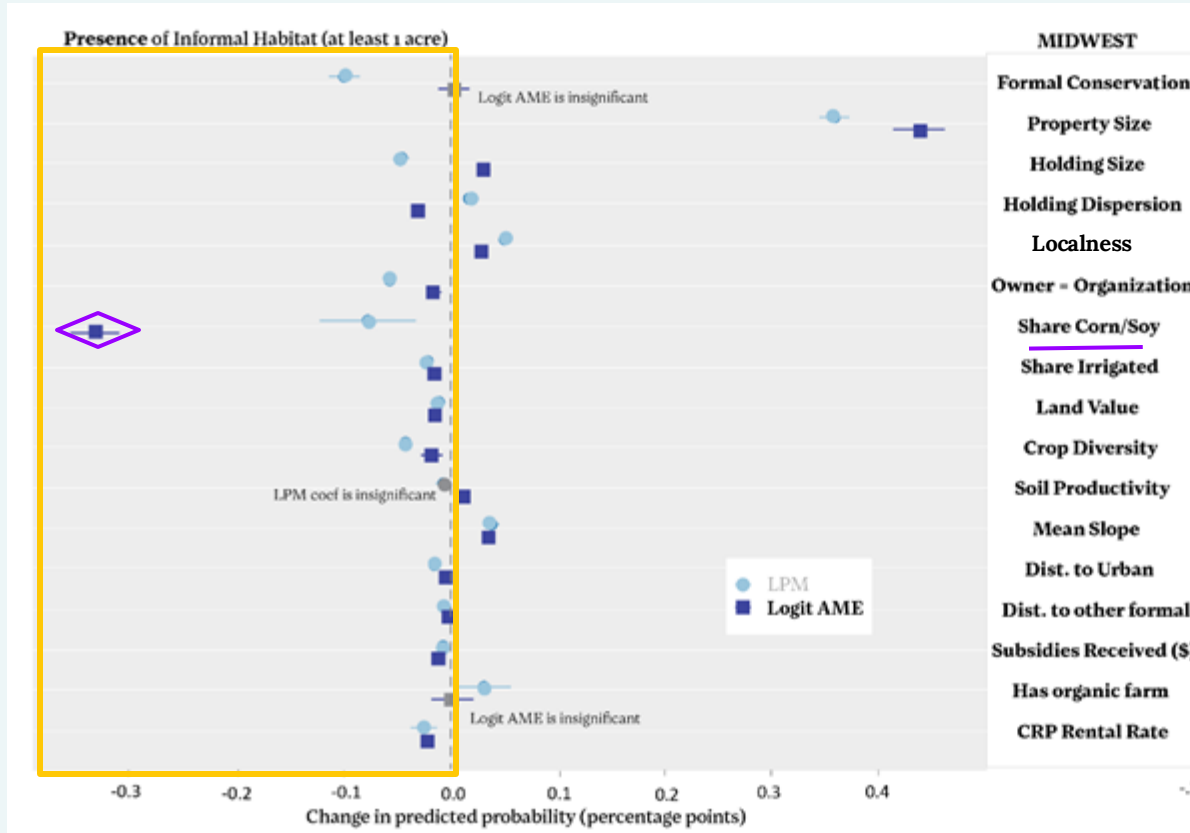
## Mean Slope

Positively  
predicts  
presence



## Corn/Soy share

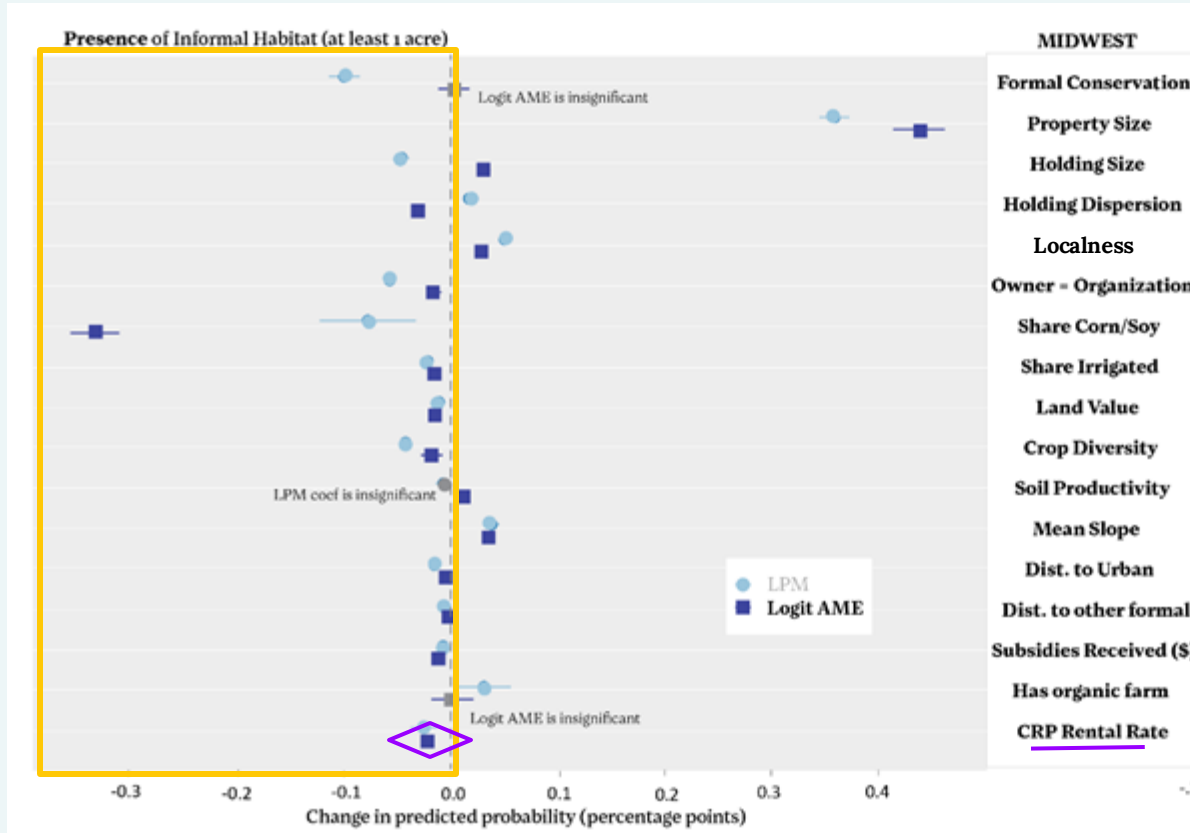
Negatively  
predicts  
presence





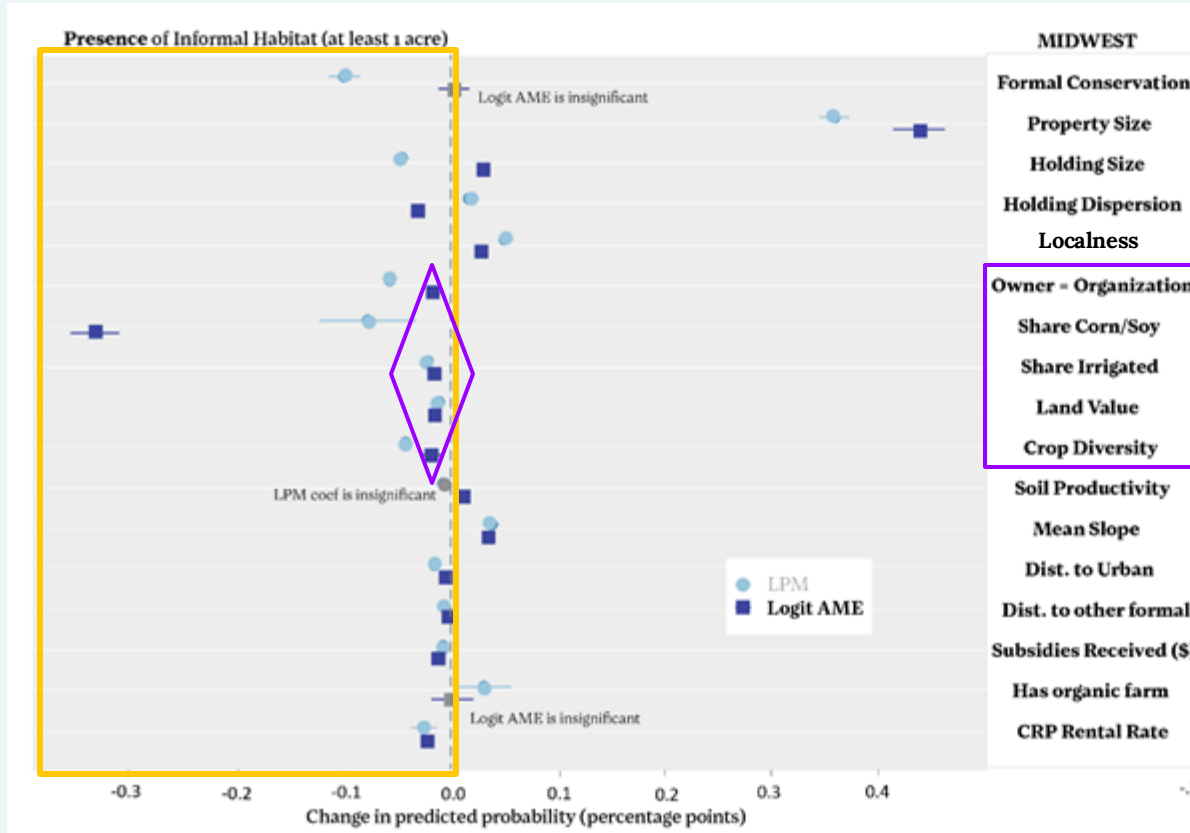
## CRP Rental Rate

Negatively  
predicts  
presence



Organizations,  
irrigation, land value,  
# of crops

Negatively  
predicts  
presence



# Predicting Presence

**Property Size**  
**Localness**  
**Mean Slope**



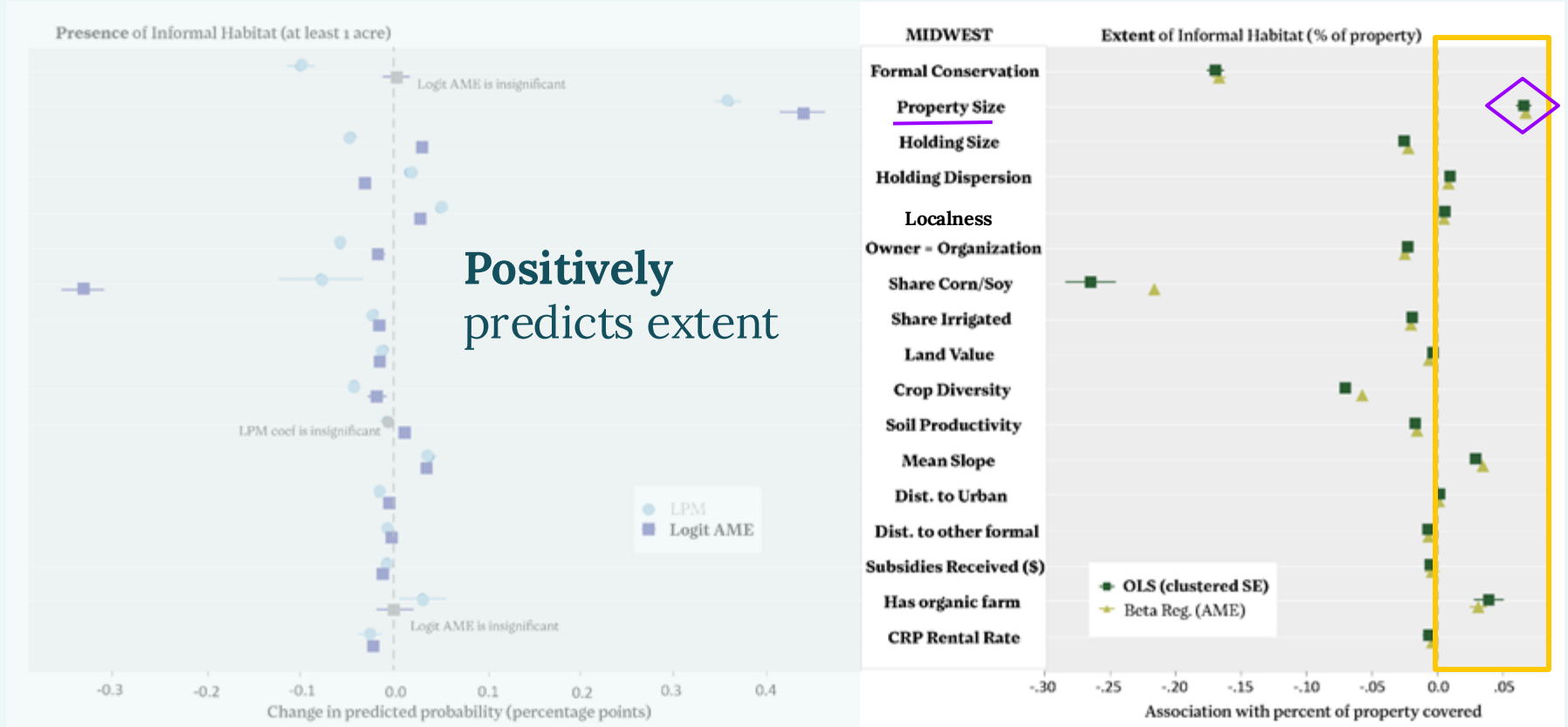
**Share Corn/Soy**  
**CRP Rental Rate**  
**Irrigation**  
**Land Value**  
**# Crops**



Of properties with **at least 1 acre of habitat**, what predicts the **extent of that habitat**?

- Percentage of property
- Conditional on already having some

# Property Size



# Organic Farm Presence

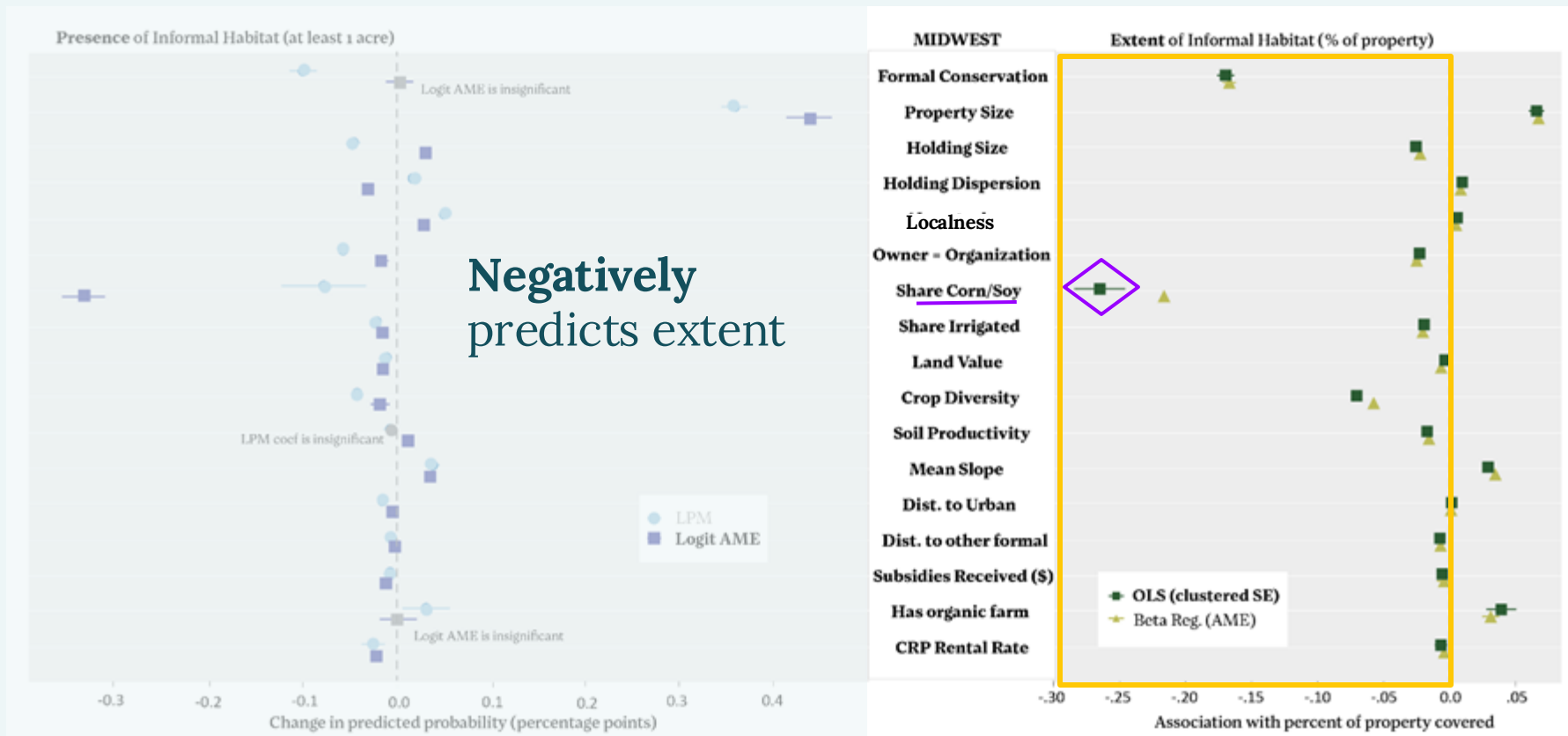


# Mean Slope

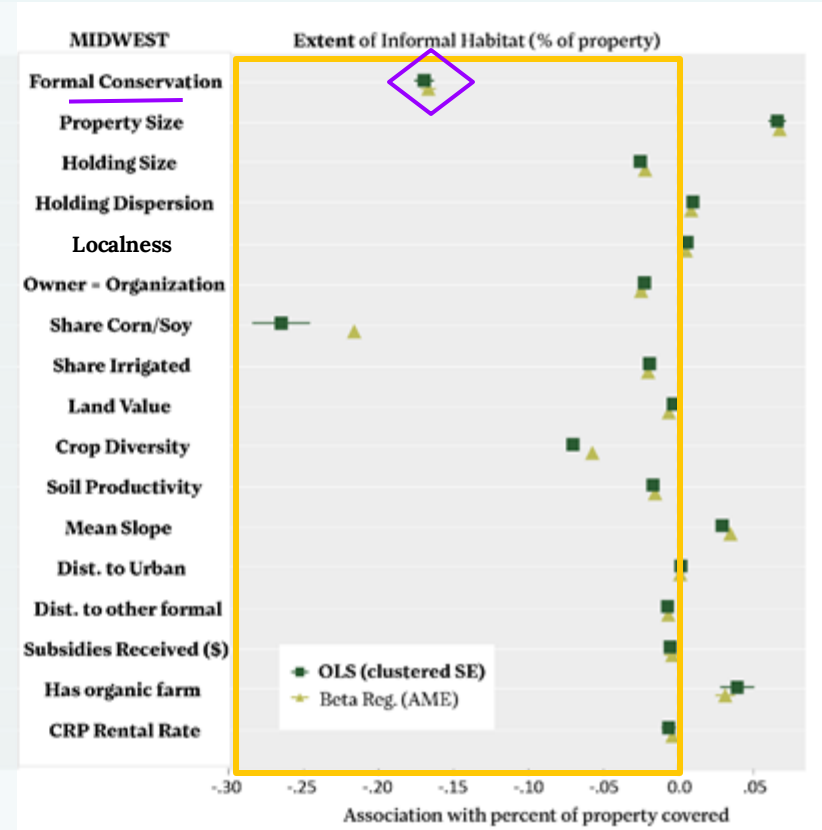
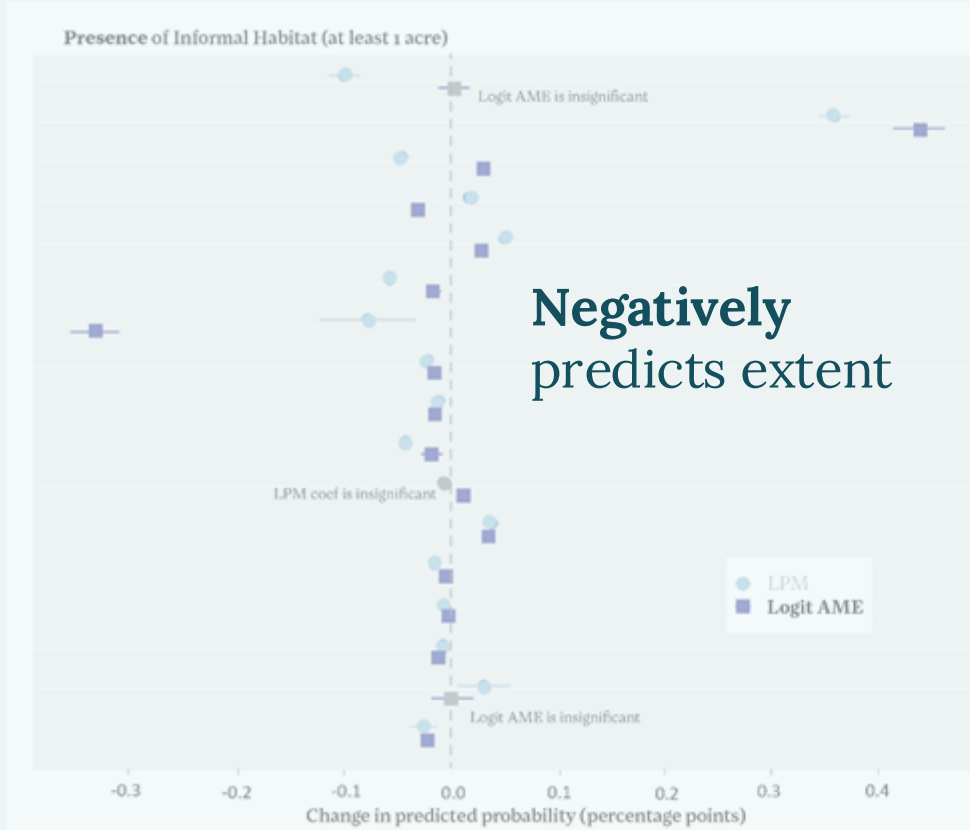




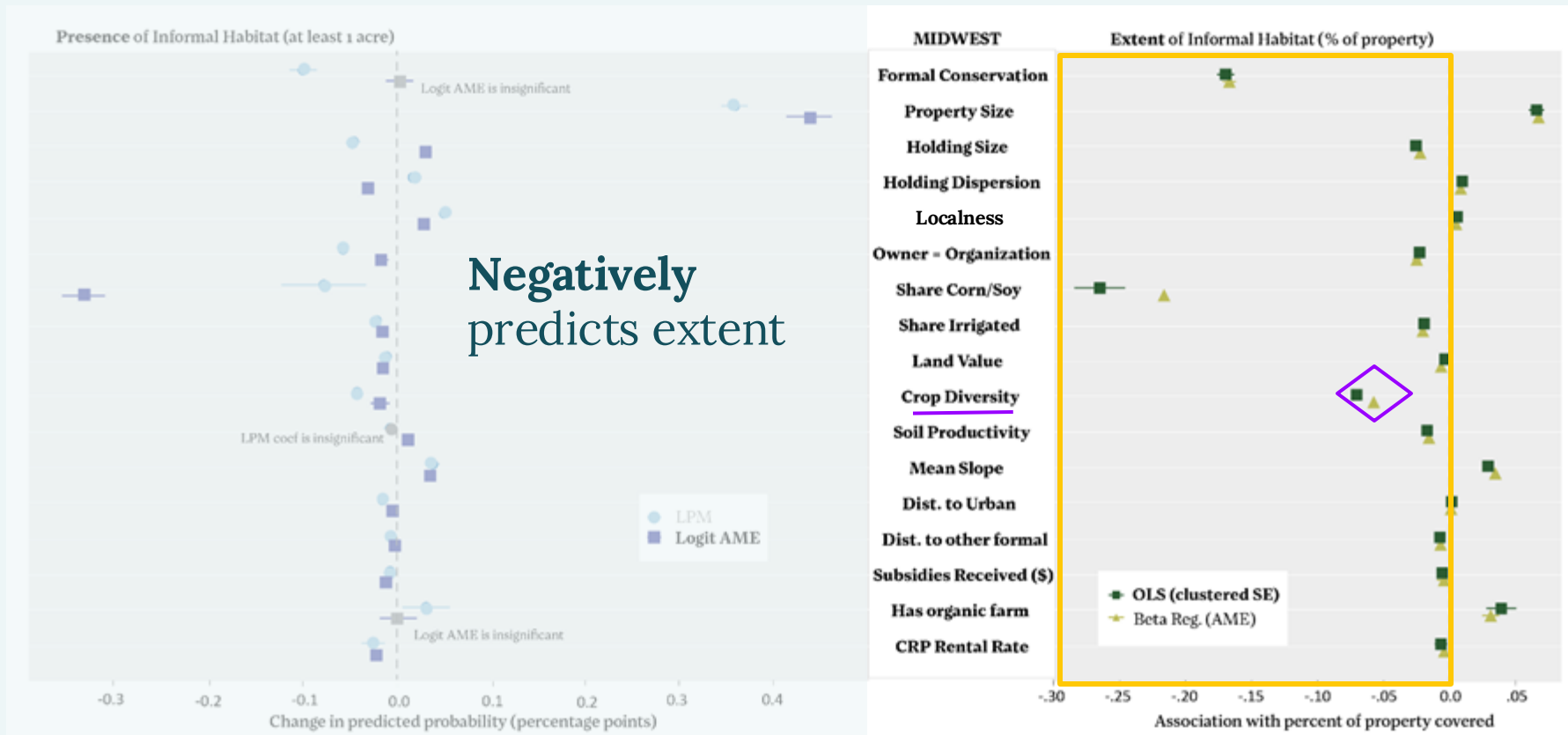
# Corn/Soy Share



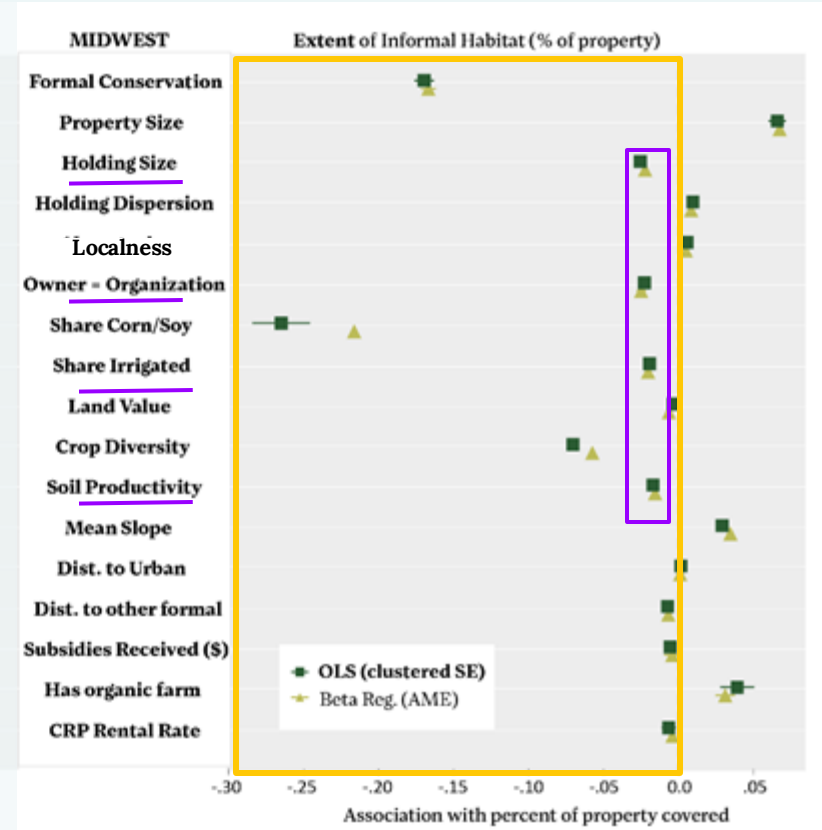
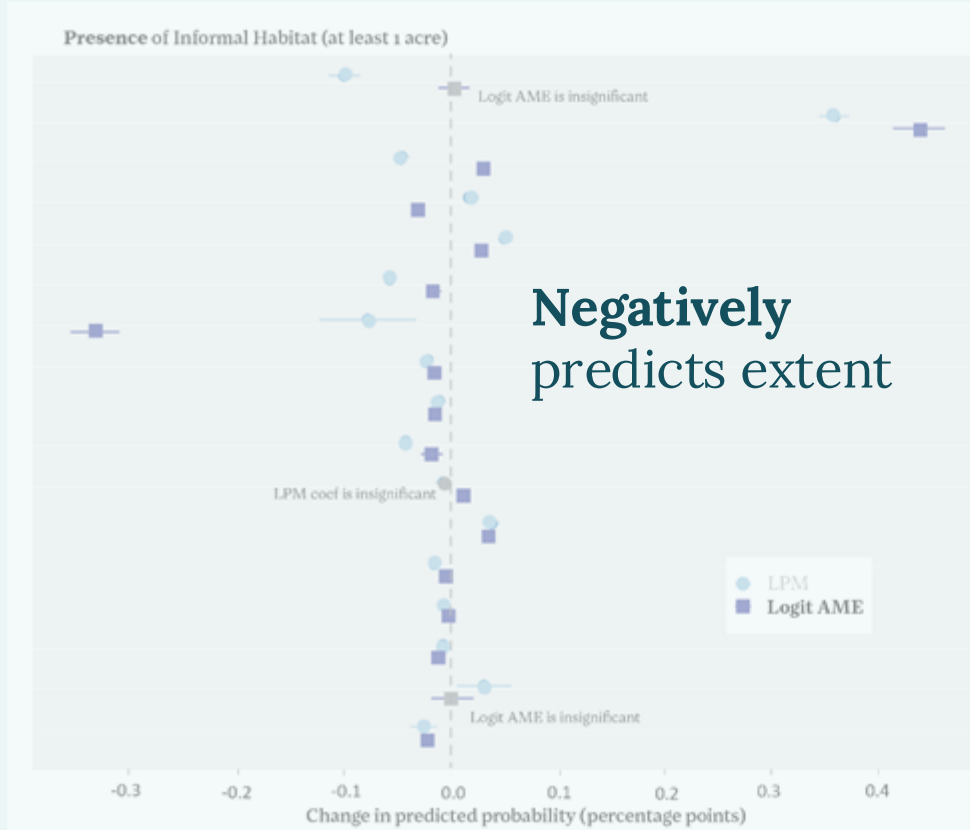
# Presence of Formal Conservation



# Number of Crops

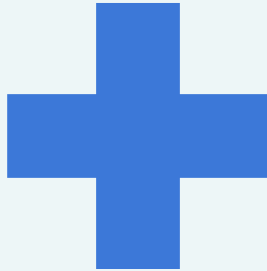


# Orgs, Irrigation, Soil Productivity, Holding Size



# Predicting Extent

**Property Size**  
**Organic Farm present**  
**Mean Slope**



**Share Corn/Soy**  
**Formal Conservation**  
**Irrigation**  
**Land Value**  
**# Crops**  
**Soil Productivity**  
**Holding Size**



# Property size: **positive** relationship with informal habitat

Presence: +43.8 points (HUGE!)

Extent: +6.8 points (modest)

- **Having vs Dedicating**

- **Larger** properties → **more likely** to have  $\geq 1$  acre of habitat
- Among properties with habitat, size → only **modest** gains in habitat **share**

- **Scale**

- Larger landholders can balance production and conservation
- Smaller properties may have to prioritize one use.

# *Informal Habitat: Embedded in Land Constraints*

**Physical constraints** (soil, slope, irrigation) associated with patterns **more** than economics

**Productive** soils → **small** habitat patches; **marginal** soils → **large** habitat shares

**Corn/soy** and irrigation strong negative predictor

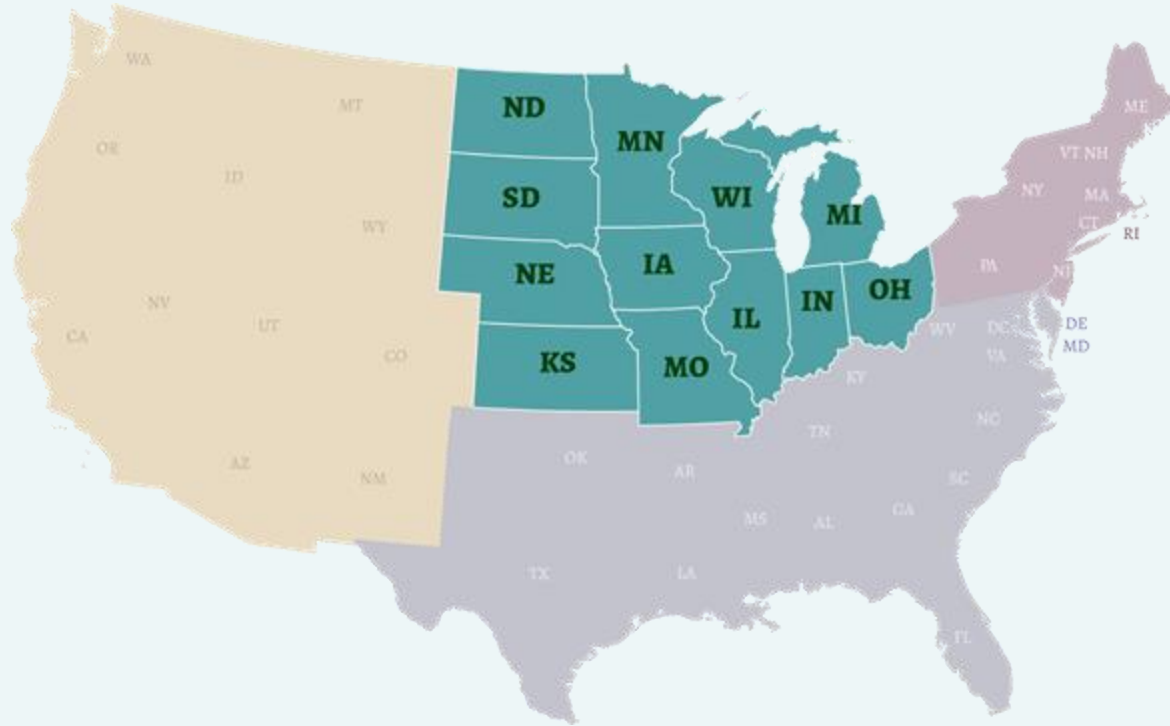
**Organic** farming shows minimal associations

Habitat concentrates where farming is *least viable*

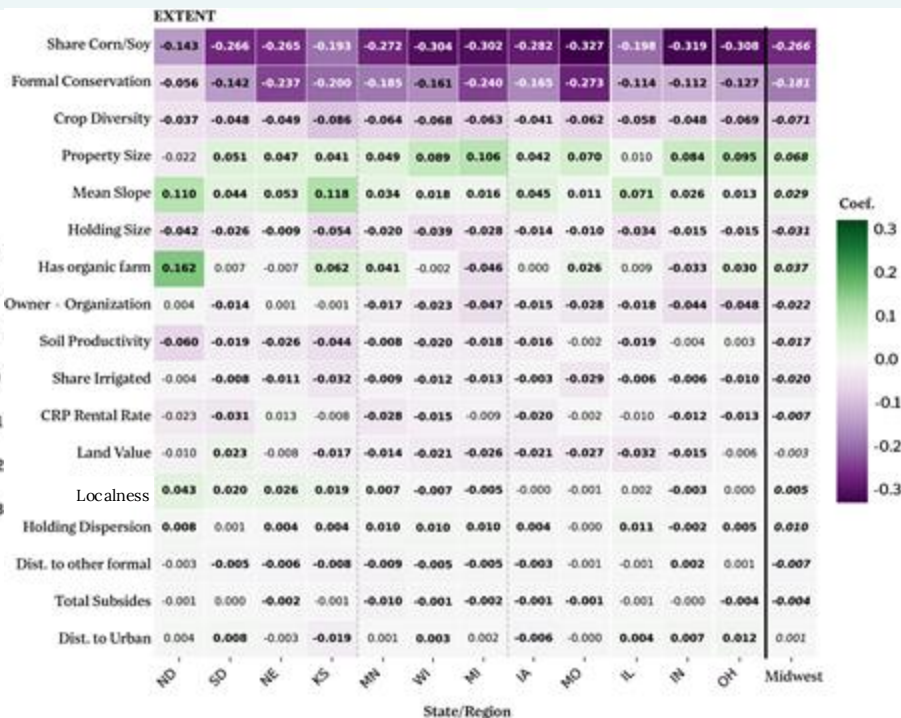
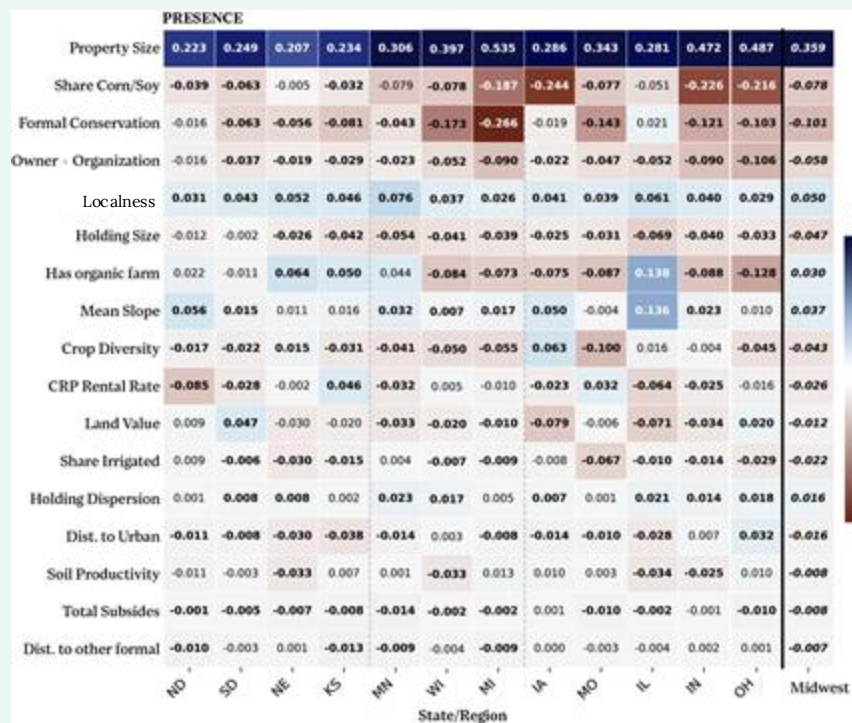




# Heterogeneity by State



# Heterogeneity by State



# Property size – positive but depends on state

- **Property size: Strongest** predictor everywhere ( $\beta = 0.21$ - $0.54$  for **presence**)
  - Largest magnitudes: Michigan, Indiana, Ohio

|                       | PRESENCE     |        |        |        |        |        |        |        |        |        |        |        |         |
|-----------------------|--------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| Property Size         | 0.223        | 0.249  | 0.207  | 0.234  | 0.306  | 0.397  | 0.535  | 0.286  | 0.343  | 0.281  | 0.472  | 0.487  | 0.359   |
| Share Corn/Soy        | -0.039       | -0.063 | -0.005 | -0.032 | -0.079 | -0.078 | -0.187 | -0.244 | -0.077 | -0.051 | -0.226 | -0.216 | -0.078  |
| Formal Conservation   | -0.016       | -0.063 | -0.056 | -0.081 | -0.043 | -0.173 | -0.266 | -0.019 | -0.143 | 0.021  | -0.121 | -0.103 | -0.101  |
| Owner - Organization  | -0.016       | -0.037 | -0.019 | -0.029 | -0.023 | -0.052 | -0.090 | -0.022 | -0.047 | -0.052 | -0.090 | -0.106 | -0.058  |
| Absenteeism           | 0.031        | 0.043  | 0.052  | 0.046  | 0.076  | 0.037  | 0.026  | 0.041  | 0.039  | 0.061  | 0.040  | 0.029  | 0.050   |
| Holding Size          | -0.012       | -0.002 | -0.026 | -0.042 | -0.054 | -0.041 | -0.039 | -0.025 | -0.031 | -0.069 | -0.040 | -0.033 | -0.047  |
| Has organic farm      | 0.022        | -0.011 | 0.064  | 0.050  | 0.044  | -0.084 | -0.073 | -0.075 | -0.087 | 0.138  | -0.088 | -0.128 | 0.030   |
| Mean Slope            | 0.056        | 0.015  | 0.011  | 0.016  | 0.032  | 0.007  | 0.017  | 0.050  | -0.004 | 0.136  | 0.023  | 0.010  | 0.037   |
| Crop Diversity        | -0.017       | -0.022 | 0.015  | -0.031 | -0.041 | -0.050 | -0.055 | 0.063  | -0.100 | 0.016  | -0.004 | -0.045 | -0.043  |
| CRP Rental Rate       | -0.085       | -0.028 | -0.002 | 0.046  | -0.032 | 0.005  | -0.010 | -0.023 | 0.032  | -0.064 | -0.025 | -0.016 | -0.026  |
| Land Value            | 0.009        | 0.047  | -0.030 | -0.020 | -0.033 | -0.020 | -0.010 | -0.079 | -0.006 | -0.071 | -0.034 | 0.020  | -0.012  |
| Share Irrigated       | 0.009        | -0.006 | -0.030 | -0.015 | 0.004  | -0.007 | -0.009 | -0.008 | -0.067 | -0.010 | -0.014 | -0.029 | -0.022  |
| Holding Dispersion    | 0.001        | 0.008  | 0.008  | 0.002  | 0.023  | 0.017  | 0.005  | 0.007  | 0.001  | 0.021  | 0.014  | 0.018  | 0.016   |
| Dist. to Urban        | -0.011       | -0.008 | -0.030 | -0.038 | -0.014 | 0.003  | -0.008 | -0.014 | -0.010 | -0.028 | 0.007  | 0.032  | -0.016  |
| Soil Productivity     | -0.011       | -0.003 | -0.033 | 0.007  | 0.001  | -0.033 | 0.013  | 0.010  | 0.003  | -0.034 | -0.025 | 0.010  | -0.008  |
| Total Subsidies       | -0.001       | -0.005 | -0.007 | -0.008 | -0.014 | -0.002 | -0.002 | 0.001  | -0.010 | -0.002 | -0.001 | -0.010 | -0.008  |
| Dist. to other formal | -0.010       | -0.003 | 0.001  | -0.013 | -0.009 | -0.004 | -0.009 | 0.000  | -0.003 | -0.004 | 0.002  | 0.001  | -0.007  |
|                       | ND           | SD     | NE     | KS     | MI     | WI     | MI     | IA     | MO     | IL     | IN     | OH     | Midwest |
|                       | State/Region |        |        |        |        |        |        |        |        |        |        |        |         |

# Those states are smaller on average

| State-Level Summary Statistics (Means)     |            |            |            |            |           |           |           |           |           |           |           |           |           |
|--------------------------------------------|------------|------------|------------|------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Variable                                   | ND         | SD         | NE         | KS         | MN        | WI        | MI        | IA        | MO        | IL        | IN        | OH        | Midwest   |
| <b>Economic</b>                            |            |            |            |            |           |           |           |           |           |           |           |           |           |
| Land Value (ln(\$/ha))                     | 8.19       | 8.57       | 8.89       | 8.75       | 9.48      | 9.80      | 9.87      | 9.77      | 9.46      | 9.72      | 9.69      | 9.82      | 9.56      |
| CRP rental rate (\$/acre/yr)               | 69.49      | 116.06     | 155.79     | 66.13      | 126.79    | 122.50    | 95.26     | 259.87    | 97.69     | 239.51    | 202.14    | 139.94    | 140.16    |
| Total subsidies (\$/prop)                  | 4583.57    | 2996.19    | 1400.84    | 1064.21    | 722.83    | 240.69    | 131.83    | 929.58    | 341.98    | 429.30    | 382.93    | 75.84     | 621.36    |
| Has formal conservation (%)                | 13.7       | 12.6       | 4.9        | 5.3        | 5.5       | 1.2       | 0.5       | 12.9      | 1.8       | 6.9       | 2.4       | 2.0       | 4.0       |
| Has organic farm (yes/no)                  | 0.001      | 0.001      | 0.001      | 0.000      | 0.001     | 0.002     | 0.001     | 0.002     | 0.000     | 0.001     | 0.001     | 0.001     | 0.001     |
| <b>Land Cover and Use</b>                  |            |            |            |            |           |           |           |           |           |           |           |           |           |
| Mean slope (degrees)                       | 2.4        | 3.0        | 3.1        | 2.3        | 2.9       | 3.6       | 2.5       | 3.5       | 5.3       | 2.6       | 3.1       | 4.1       | 3.3       |
| Crop Diversity ( of crops)                 | 2.6        | 1.4        | 1.4        | 1.5        | 1.0       | 0.8       | 0.7       | 1.3       | 0.4       | 1.1       | 0.8       | 0.5       | 0.91      |
| Share of Corn/Soy cover                    | 18         | 23         | 34         | 14         | 17        | 08        | 05        | 41        | 07        | 34        | 19        | 11        | 16        |
| Irrigated Share (%)                        | 02         | 03         | 36         | 10         | 01        | 01        | 01        | 01        | 03        | 02        | 02        | 00        | 03        |
| Soil Productivity (NCCPI 0-1000)           | 423        | 450        | 577        | 578        | 522       | 474       | 470       | 748       | 511       | 701       | 631       | 596       | 550       |
| <b>Ownership</b>                           |            |            |            |            |           |           |           |           |           |           |           |           |           |
| Absenteeism (0=absentee, 1=local)          | 0.35       | 0.37       | 0.49       | 0.56       | 0.55      | 0.60      | 0.60      | 0.51      | 0.59      | 0.58      | 0.73      | 0.65      | 0.56      |
| Owner is Organization (yes/no)             | 0.05       | 0.08       | 0.13       | 0.10       | 0.05      | 0.06      | 0.04      | 0.11      | 0.06      | 0.07      | 0.07      | 0.06      | 0.07      |
| <b>Property area (ac)</b>                  | <b>173</b> | <b>184</b> | <b>180</b> | <b>139</b> | <b>56</b> | <b>30</b> | <b>19</b> | <b>74</b> | <b>47</b> | <b>48</b> | <b>27</b> | <b>21</b> | <b>54</b> |
| Holding area (ac)                          | 37,202     | 13,293     | 280,129    | 33,468     | 43,321    | 85,593    | 66,984    | 116,313   | 20,078    | 86,509    | 140,660   | 217,951   | 97,076    |
| Holding dispersion (km)                    | 71         | 61         | 88         | 65         | 64        | 68        | 47        | 61        | 38        | 59        | 40        | 43        | 54        |
| <b>Location</b>                            |            |            |            |            |           |           |           |           |           |           |           |           |           |
| Distance to closest enrolled practice (km) | 416        | 386        | 424        | 516        | 306       | 238       | 240       | 316       | 410       | 385       | 488       | 561       | 377       |
| Distance to urban area (km)                | 7.8        | 7.3        | 5.254      | 5.1        | 5.4       | 4.5       | 4.5       | 3.7       | 4.5       | 3.2       | 2.7       | 3.0       | 4.3       |
| Mean % Informal Habitat                    | 48         | 59         | 50         | 61         | 66        | 74        | 74        | 47        | 81        | 53        | 64        | 71        | 66        |

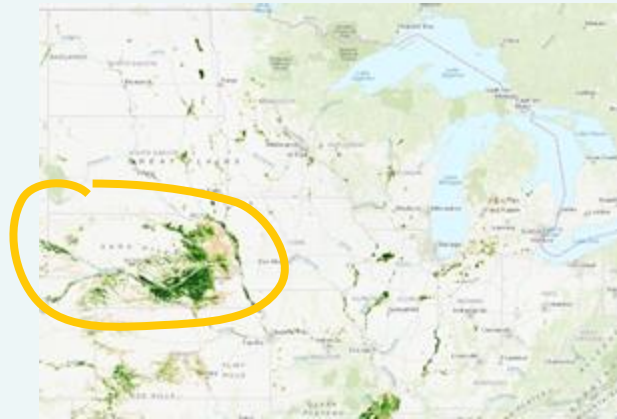
# Formal conservation presence

## Negatively predicts informal extent

- **Most negative** in NE ( $\beta = -0.24$ ), MO ( $\beta = -0.27$ )

|                            | EXTENT        |               |               |               |               |               |               |               |               |               |               |               |               |
|----------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Share Corn/Soy             | -0.143        | -0.266        | -0.165        | -0.193        | -0.272        | -0.304        | -0.302        | -0.282        | -0.127        | -0.198        | -0.319        | -0.308        | -0.266        |
| <b>Formal Conservation</b> | <b>-0.056</b> | <b>-0.142</b> | <b>-0.237</b> | <b>-0.200</b> | <b>-0.185</b> | <b>-0.161</b> | <b>-0.240</b> | <b>-0.165</b> | <b>-0.273</b> | <b>-0.114</b> | <b>-0.112</b> | <b>-0.127</b> | <b>-0.181</b> |
| Crop Diversity             | -0.037        | -0.048        | -0.049        | -0.086        | -0.064        | -0.068        | -0.063        | -0.041        | -0.062        | -0.058        | -0.048        | -0.069        | -0.071        |
| Property Size              | -0.022        | 0.051         | 0.047         | 0.041         | 0.049         | 0.089         | 0.106         | 0.042         | 0.070         | 0.010         | 0.084         | 0.095         | 0.068         |
| Mean Slope                 | 0.110         | 0.044         | 0.053         | 0.118         | 0.034         | 0.018         | 0.016         | 0.045         | 0.011         | 0.071         | 0.026         | 0.013         | 0.029         |
| Holding Size               | -0.042        | -0.026        | -0.009        | -0.054        | -0.020        | -0.039        | -0.028        | -0.014        | -0.010        | -0.034        | -0.015        | -0.015        | -0.031        |
| Has organic farm           | 0.162         | 0.007         | -0.007        | 0.062         | 0.041         | -0.002        | -0.046        | 0.000         | 0.026         | 0.009         | -0.033        | 0.030         | 0.037         |
| Owner - Organization       | 0.004         | -0.014        | 0.001         | -0.001        | -0.017        | -0.023        | -0.047        | -0.015        | -0.028        | -0.018        | -0.044        | -0.048        | -0.022        |
| Soil Productivity          | -0.060        | -0.019        | -0.026        | -0.044        | -0.008        | -0.020        | -0.018        | -0.016        | -0.002        | -0.019        | -0.004        | 0.003         | -0.017        |
| Share Irrigated            | -0.004        | -0.008        | -0.011        | -0.032        | -0.009        | -0.012        | -0.013        | -0.003        | -0.029        | -0.006        | -0.006        | -0.010        | -0.020        |
| CRP Rental Rate            | -0.023        | -0.031        | 0.013         | -0.008        | -0.028        | -0.015        | -0.009        | -0.020        | -0.002        | -0.010        | -0.012        | -0.013        | -0.007        |
| Land Value                 | -0.010        | 0.023         | -0.008        | -0.017        | -0.014        | -0.021        | -0.026        | -0.021        | -0.027        | -0.032        | -0.015        | -0.006        | -0.003        |
| Absenteeism                | 0.043         | 0.020         | 0.026         | 0.019         | 0.007         | -0.007        | -0.005        | -0.000        | -0.001        | 0.002         | -0.003        | 0.000         | 0.005         |
| Holding Dispersion         | 0.008         | 0.001         | 0.004         | 0.004         | 0.010         | 0.010         | 0.010         | 0.004         | -0.000        | 0.011         | -0.002        | 0.005         | 0.010         |
| Dist. to other formal      | -0.003        | -0.005        | -0.006        | -0.008        | -0.009        | -0.005        | -0.005        | -0.003        | -0.001        | -0.001        | 0.002         | 0.001         | -0.007        |
| Total Subsidies            | -0.001        | 0.000         | -0.002        | -0.001        | -0.010        | -0.001        | -0.002        | -0.001        | -0.001        | -0.001        | -0.000        | -0.004        | -0.004        |
| Dist. to Urban             | 0.004         | 0.008         | -0.003        | -0.019        | 0.001         | 0.003         | 0.002         | -0.006        | -0.000        | 0.004         | 0.007         | 0.012         | 0.001         |
|                            | ND            | SD            | NE            | KS            | MN            | WI            | MI            | IA            | MO            | IL            | IN            | OH            | Midwest       |
|                            |               |               |               |               |               |               |               |               |               |               |               |               |               |

# Nebraska – way more irrigation



Level Summary Statistics (Means)

|                                                                                   |                             | SD      | NE      | KS      | MN      | WI     | MI     | IA      | MO     | IL     | IN      | OH      | Midwest |        |
|-----------------------------------------------------------------------------------|-----------------------------|---------|---------|---------|---------|--------|--------|---------|--------|--------|---------|---------|---------|--------|
|  |                             | 8.57    | 8.89    | 8.75    | 9.48    | 9.80   | 9.87   | 9.77    | 9.46   | 9.72   | 9.69    | 9.82    | 9.56    |        |
|                                                                                   |                             | 116.06  | 155.79  | 66.13   | 126.79  | 122.50 | 95.26  | 259.87  | 97.69  | 239.51 | 202.14  | 139.94  | 140.16  |        |
|                                                                                   | Total subsidies (\$/prop)   | 4583.57 | 2996.19 | 1400.84 | 1064.21 | 722.83 | 240.69 | 131.83  | 929.58 | 341.98 | 429.30  | 382.93  | 75.84   | 621.36 |
|                                                                                   | Has formal conservation (%) | 13.7    | 12.6    | 4.9     | 5.3     | 5.5    | 1.2    | 0.5     | 12.9   | 1.8    | 6.9     | 2.4     | 2.0     | 4.0    |
| Has organic farm (yes/no)                                                         | 0.001                       | 0.001   | 0.001   | 0.000   | 0.001   | 0.002  | 0.001  | 0.002   | 0.000  | 0.001  | 0.001   | 0.001   | 0.001   |        |
| Land Cover and Use                                                                |                             |         |         |         |         |        |        |         |        |        |         |         |         |        |
| Mean slope (degrees)                                                              | 2.4                         | 3.0     | 3.1     | 2.3     | 2.9     | 3.6    | 2.5    | 3.5     | 5.3    | 2.6    | 3.1     | 4.1     | 3.3     |        |
| Crop Diversity ( # of crops)                                                      | 2.6                         | 1.4     | 1.4     | 1.5     | 1.0     | 0.8    | 0.7    | 1.3     | 0.4    | 1.1    | 0.8     | 0.5     | 0.91    |        |
| Share of Corn/Soy cover                                                           | 18                          | 23      | 34      | 14      | 17      | 08     | 05     | 41      | 07     | 34     | 19      | 11      | 16      |        |
| Irrigated Share (%)                                                               | 02                          | 03      | 36      | 10      | 01      | 01     | 01     | 01      | 03     | 02     | 02      | 00      | 03      |        |
| Soil Productivity (NCCPI 0-1000)                                                  | 423                         | 450     | 577     | 578     | 522     | 474    | 470    | 748     | 511    | 701    | 631     | 596     | 550     |        |
| Ownership                                                                         |                             |         |         |         |         |        |        |         |        |        |         |         |         |        |
| Absenteeism (0=absentee, 1=local)                                                 | 0.35                        | 0.37    | 0.49    | 0.56    | 0.55    | 0.60   | 0.60   | 0.51    | 0.59   | 0.58   | 0.73    | 0.65    | 0.56    |        |
| Owner is Organization (yes/no)                                                    | 0.05                        | 0.08    | 0.13    | 0.10    | 0.05    | 0.06   | 0.04   | 0.11    | 0.06   | 0.07   | 0.07    | 0.06    | 0.07    |        |
| Property area (ac)                                                                | 173                         | 184     | 180     | 139     | 56      | 30     | 19     | 74      | 47     | 48     | 27      | 21      | 54      |        |
| Holding area (ac)                                                                 | 37,202                      | 13,295  | 280,129 | 33,468  | 43,321  | 85,593 | 66,984 | 116,313 | 20,078 | 86,509 | 140,660 | 217,951 | 97,076  |        |
| Holding dispersion (km)                                                           | 71                          | 61      | 88      | 65      | 64      | 68     | 47     | 61      | 38     | 59     | 40      | 43      | 54      |        |
| Location                                                                          |                             |         |         |         |         |        |        |         |        |        |         |         |         |        |
| Distance to closest enrolled practice (km)                                        | 416                         | 386     | 424     | 516     | 306     | 238    | 240    | 316     | 410    | 385    | 488     | 561     | 377     |        |
| Distance to urban area (km)                                                       | 7.8                         | 7.3     | 5.254   | 5.1     | 5.4     | 4.5    | 4.5    | 3.7     | 4.5    | 3.2    | 2.7     | 3.0     | 4.3     |        |
| Mean % Informal Habitat                                                           | 48                          | 59      | 50      | 61      | 66      | 74     | 74     | 47      | 81     | 53     | 64      | 71      | 66      |        |



# Local Ownership: Positive in Northern Plains

| EXTENT                |        |        |        |        |        |        |        |        |        |        |        |        |         |
|-----------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| Share Corn/Soy        | -0.143 | -0.266 | -0.265 | -0.193 | -0.272 | -0.304 | -0.302 | -0.282 | -0.327 | -0.198 | -0.319 | -0.308 | -0.266  |
| Formal Conservation   | -0.056 | -0.142 | -0.237 | -0.200 | -0.185 | -0.161 | -0.240 | -0.165 | -0.273 | -0.114 | -0.112 | -0.127 | -0.183  |
| Crop Diversity        | -0.037 | -0.048 | -0.049 | -0.086 | -0.064 | -0.068 | -0.063 | -0.041 | -0.062 | -0.058 | -0.048 | -0.069 | -0.071  |
| Property Size         | -0.022 | 0.051  | 0.047  | 0.041  | 0.049  | 0.089  | 0.106  | 0.042  | 0.070  | 0.010  | 0.084  | 0.095  | 0.068   |
| Mean Slope            | 0.110  | 0.044  | 0.053  | 0.118  | 0.034  | 0.018  | 0.016  | 0.045  | 0.011  | 0.071  | 0.026  | 0.013  | 0.029   |
| Holding Size          | -0.042 | -0.026 | -0.009 | -0.054 | -0.020 | -0.039 | -0.028 | -0.014 | -0.010 | -0.034 | -0.015 | -0.015 | -0.031  |
| Has organic farm      | 0.162  | 0.007  | -0.007 | 0.062  | 0.041  | -0.002 | -0.046 | 0.000  | 0.026  | 0.009  | -0.033 | 0.030  | 0.037   |
| Owner - Organization  | 0.004  | -0.014 | 0.001  | -0.001 | -0.017 | -0.023 | -0.047 | -0.015 | -0.028 | -0.018 | -0.044 | -0.048 | -0.022  |
| Soil Productivity     | -0.060 | -0.019 | -0.026 | -0.044 | -0.008 | -0.020 | -0.018 | -0.016 | -0.002 | -0.019 | -0.004 | 0.003  | -0.017  |
| Share Irrigated       | -0.004 | -0.008 | -0.011 | -0.032 | -0.009 | -0.012 | -0.013 | -0.003 | -0.029 | -0.006 | -0.006 | -0.010 | -0.020  |
| CRP Rental Rate       | -0.023 | -0.031 | 0.013  | -0.008 | -0.028 | -0.015 | -0.009 | -0.020 | -0.002 | -0.010 | -0.012 | -0.013 | -0.007  |
| Land Value            | 0.023  | 0.023  | -0.008 | -0.017 | -0.014 | -0.021 | -0.026 | -0.021 | -0.027 | -0.032 | -0.015 | -0.006 | -0.003  |
| Localness             | 0.043  | 0.020  | 0.026  | 0.019  | 0.007  | -0.007 | -0.005 | -0.000 | -0.001 | 0.002  | -0.003 | 0.000  | 0.005   |
| Holding Dispersion    | 0.008  | 0.001  | 0.004  | 0.004  | 0.010  | 0.010  | 0.010  | 0.004  | -0.000 | 0.011  | -0.002 | 0.005  | 0.010   |
| Dist. to other formal | -0.003 | -0.005 | -0.006 | -0.008 | -0.009 | -0.005 | -0.005 | -0.003 | -0.001 | -0.001 | 0.002  | 0.001  | -0.007  |
| Total Subsidies       | -0.001 | 0.000  | -0.002 | -0.001 | -0.010 | -0.001 | -0.002 | -0.001 | -0.001 | -0.001 | -0.000 | -0.004 | -0.004  |
| Dist. to Urban        | 0.004  | 0.008  | -0.003 | -0.019 | 0.001  | 0.003  | 0.002  | -0.006 | -0.000 | 0.004  | 0.007  | 0.012  | 0.001   |
|                       | ND     | SD     | NE     | KS     | MN     | WI     | MI     | IA     | MO     | IL     | IN     | OH     | Midwest |
| State/Region          |        |        |        |        |        |        |        |        |        |        |        |        |         |

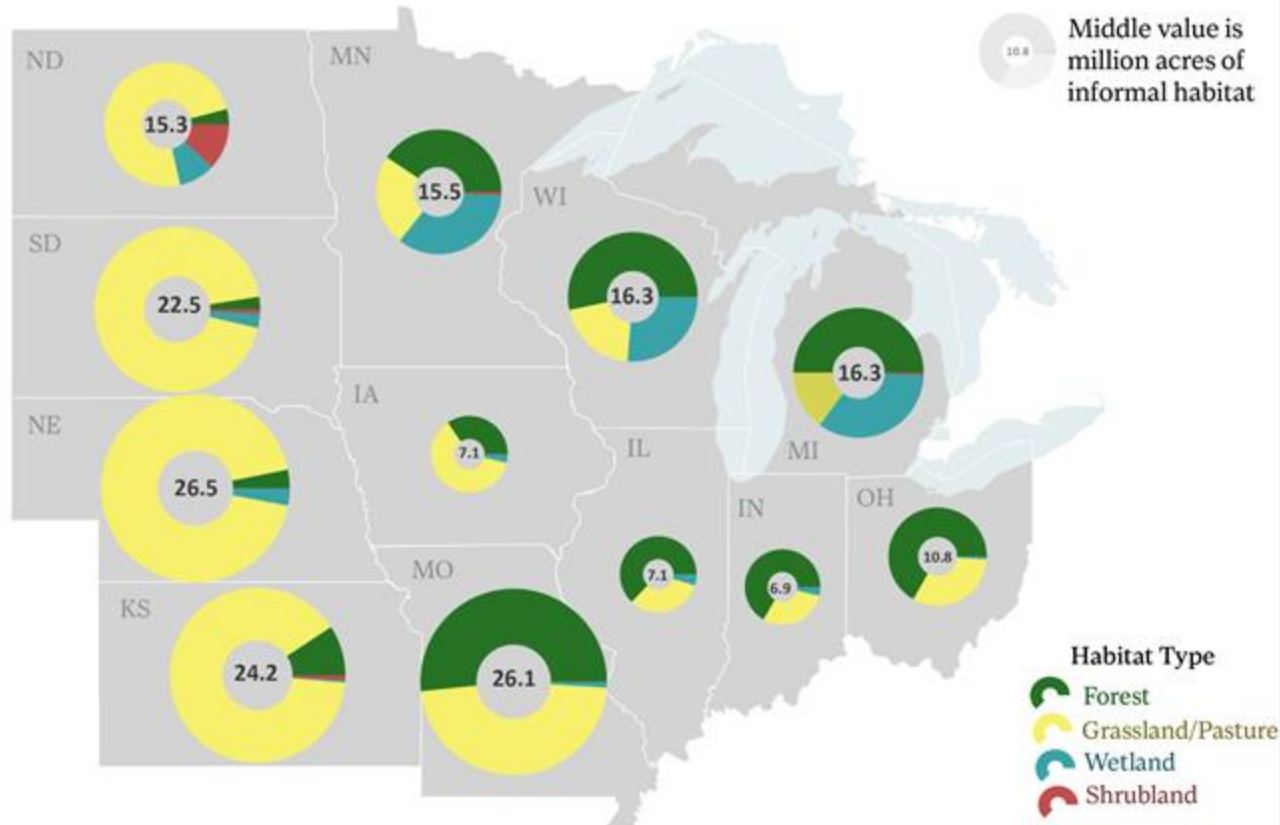
# Heterogeneity by Habitat Type



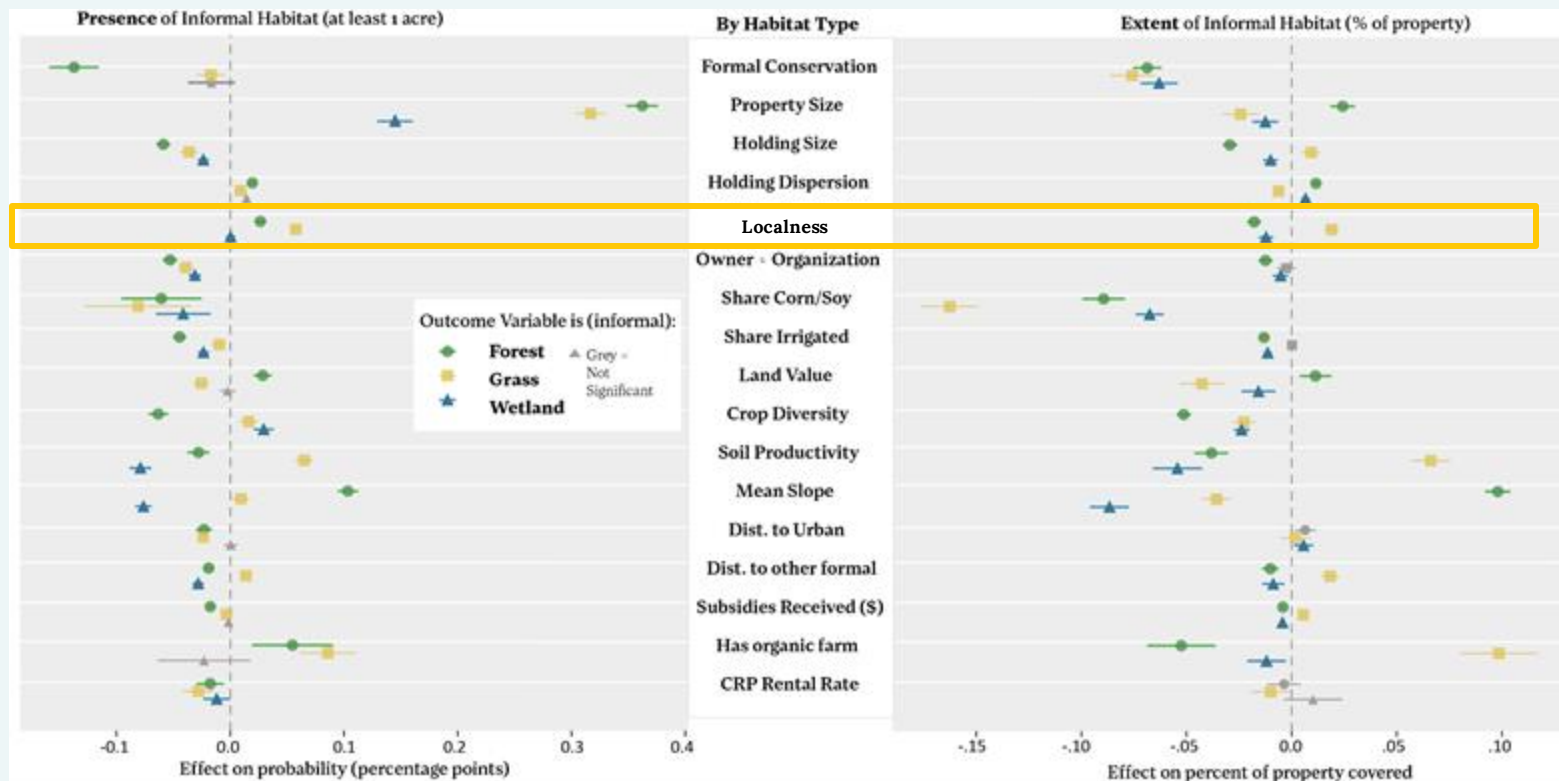


# Informal Habitat Distribution

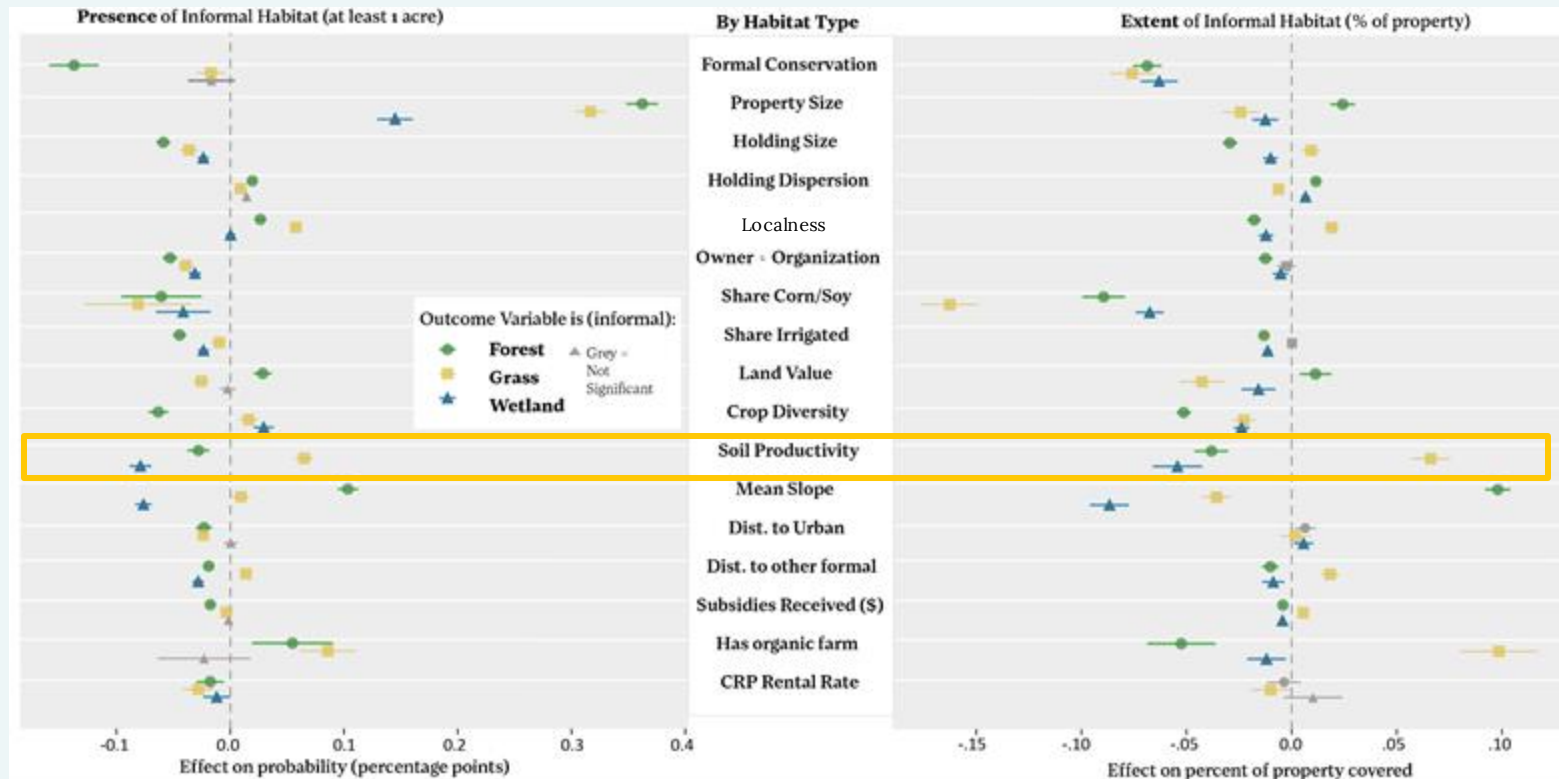
Midwest Informal Habitat Composition by State  
on rural, private properties



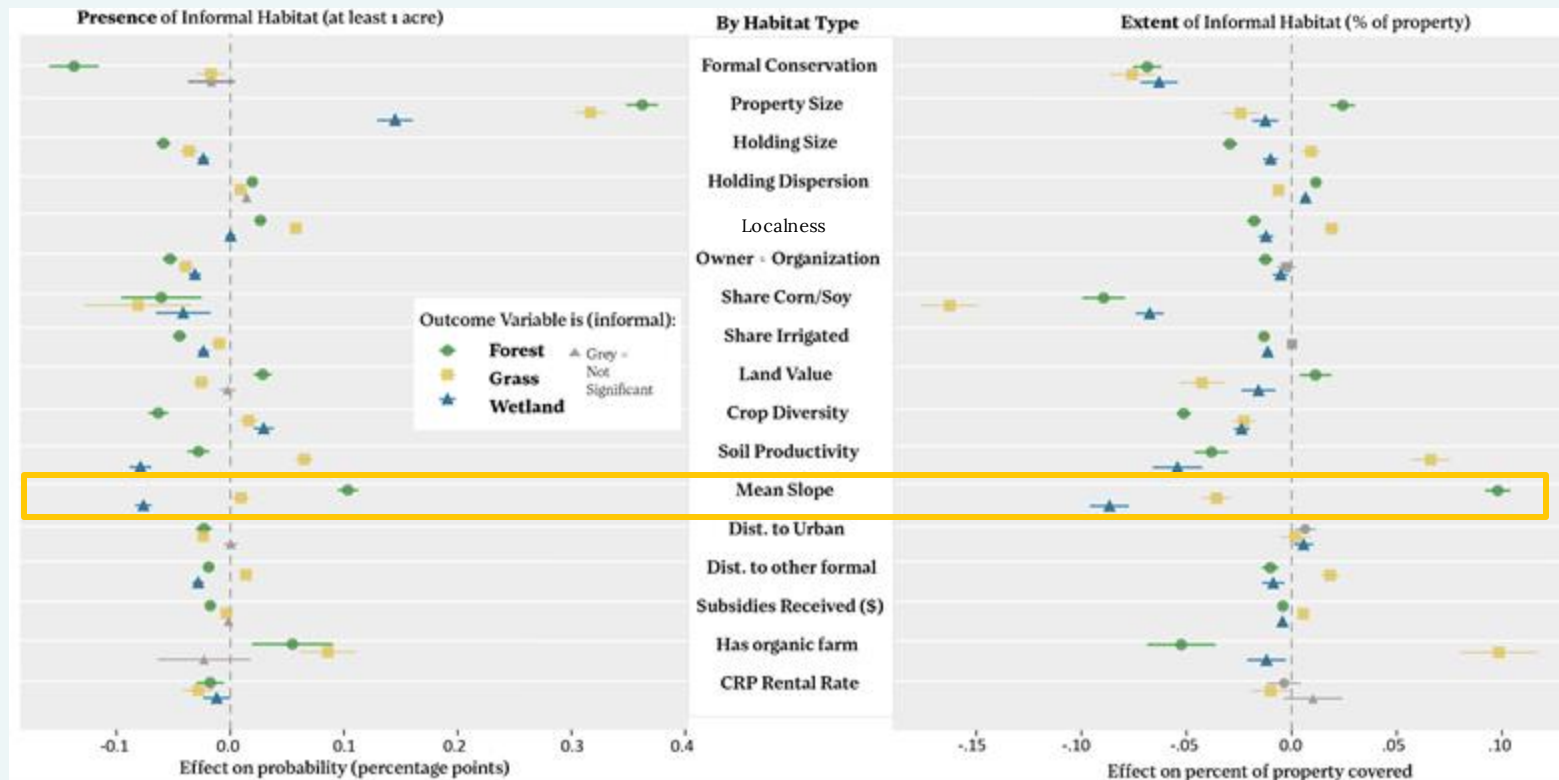
# More Local → More Grass



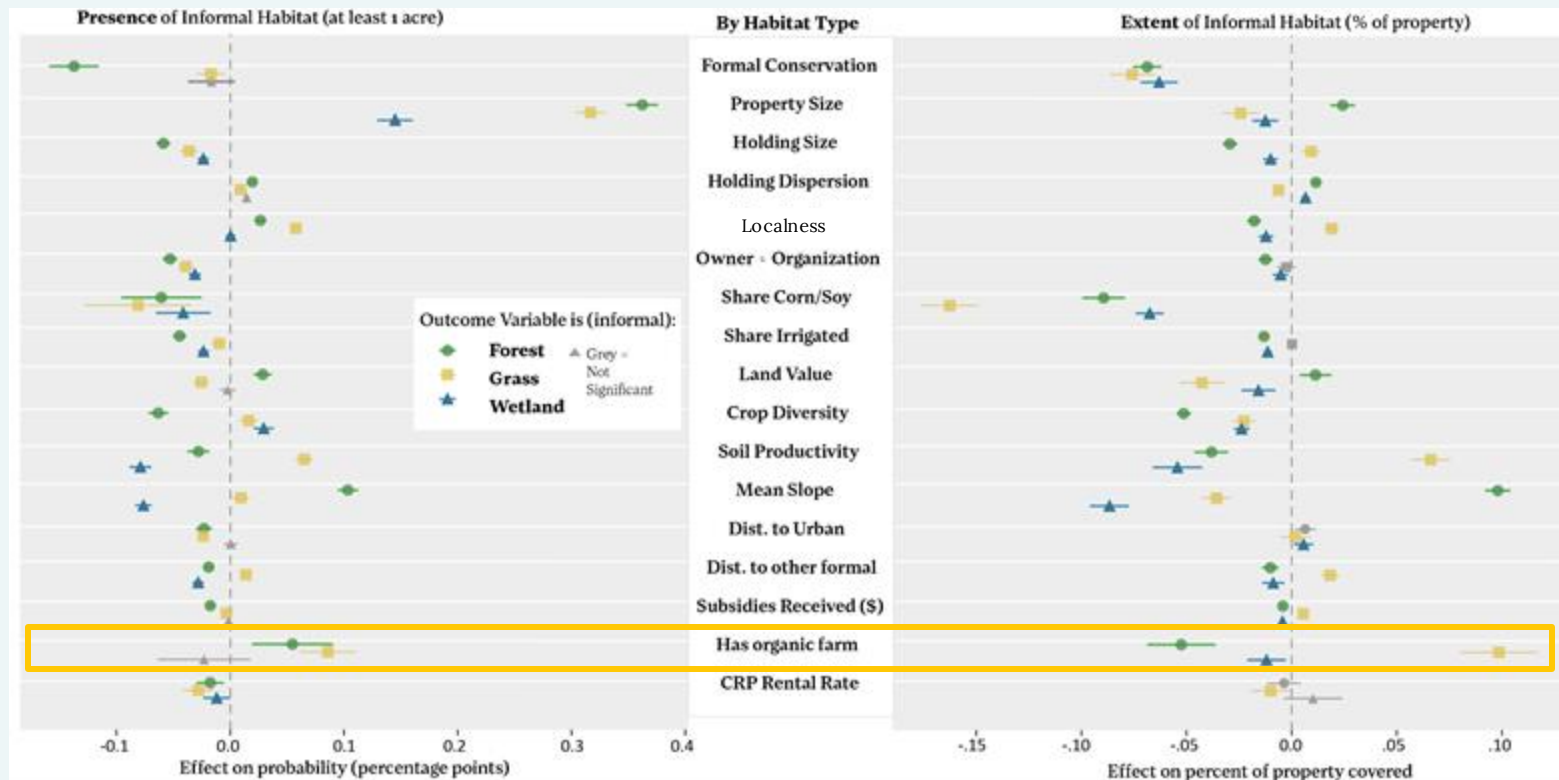
# More Productive Soils → More Grass



# Steeper Slopes → More Forest



# Organic Farm → More grass



# Organic Farm → More grass



# Different Habitats Need Different Conservation Approaches

- **Forests:** Persist on steep, less productive land; may have amenity/timber value
- **Grasslands:** Associated with extensive grazing and local ownership
- **Wetlands:** Most constrained by topography; weaker management associations
- Uniform conservation policies can't effectively address ecological and economic heterogeneity across habitat types

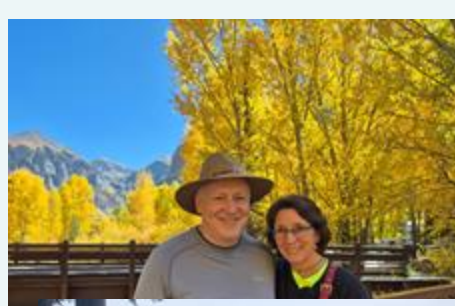
# The Vast Extent of Private Lands Conservation

- **195 million acres** of informal habitat
- **Regional complexity** matters
  - North Dakota ≠ Ohio
- Structural **vulnerability...stability?**
- Formal programs get attention & funding
  
- **Path forward**
  - Understand what exists, who maintains it, what threatens it
  - Support existing stewardship alongside formal programs



# Research Agenda – Now to the PhD!!

- **National assessment:** How does informal habitat vary across regions?
- **Causal analysis:** How does it respond to price changes, policy shifts?
- **Landowner interviews:** What drives decision-making?
- **Ecological value:** Biodiversity, carbon, water quality, connectivity
- **Urban/suburban:** Lawns, gardens, residential properties
  - NAIP imagery, machine learning



# Thank you!!!!!!!



**Holly Gibbs**  
PROFESSOR AND DIRECTOR OF  
GLUE LAB



**Tyler Lark**  
LEAD SCIENTIST



**Matt Christie**  
LEAD SCIENTIST



**Lisa Rausch**  
LEAD SCIENTIST



**Seth Spawn**  
ALUMNI



**Joe App**  
RESEARCH STAFF



**Tara Mittelberg**  
PHD STUDENT



**Malena Candino**  
RESEARCH STAFF



**Marin Elisabeth Skidmore**  
ALUMNI



**Matt Bougie**  
RESEARCH STAFF



**Raines Lucas**  
UNDERGRADUATE STUDENT



**Sam Kuemmel**  
UNDERGRADUATE STUDENT



**Lucas Palomino**  
RESEARCH STAFF



**Howard Veregin**  
Wisconsin State Cartographer  
[www.dnr.wisconsin.gov](http://www.dnr.wisconsin.gov)  
Wisconsin State Cartographer's Office  
(608) 262-6852  
Room 384 Science Hall

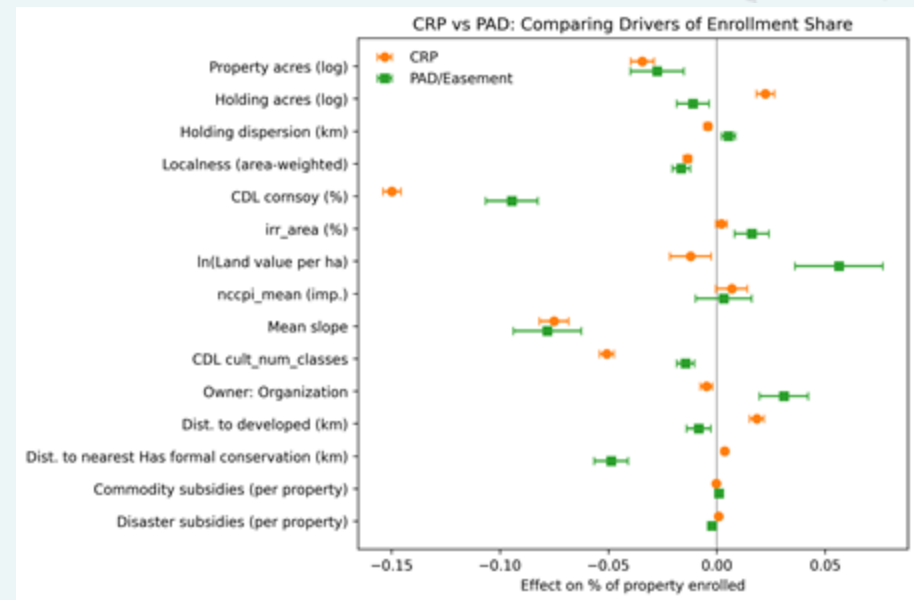
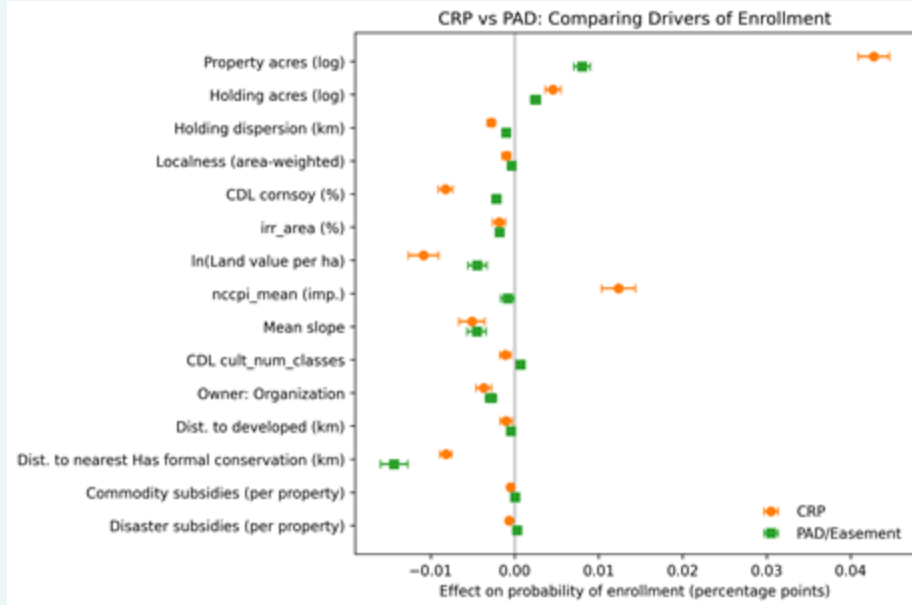
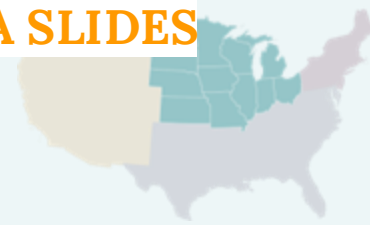


**Andrew Stevens**  
Associate Professor  
[astevens@wisc.edu](mailto:astevens@wisc.edu)  
(608) 263-3212  
330 Taylor Hall  
427 Lorch St  
Madison, WI 53706  
Interests  
Agricultural and food policy, land  
Office Hours  
On substantial, 2025-26 academic

**GEOG901: Seminar in Cultural Ge...**  
**FA25 GEOG 901 001**  
**Fall 2025-2026**

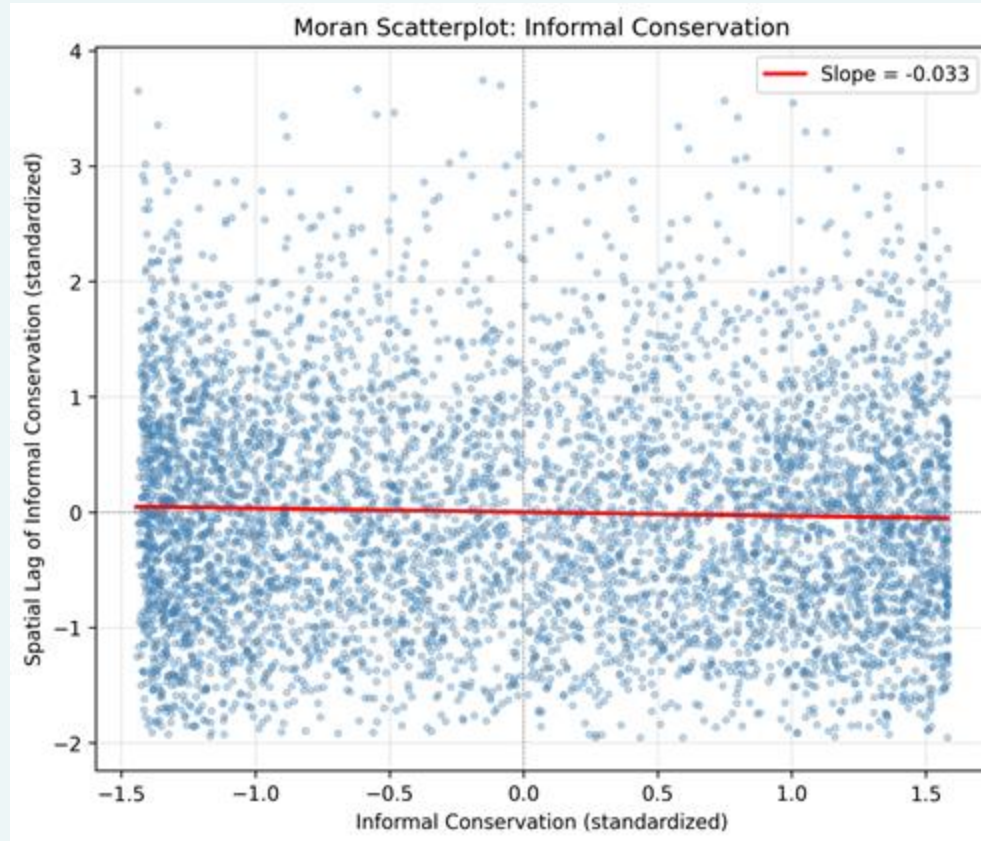
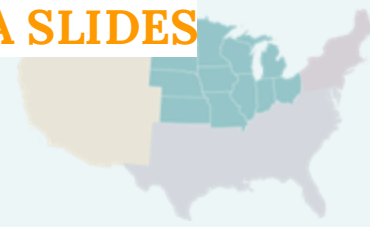


# Results: CRP Vs Easements

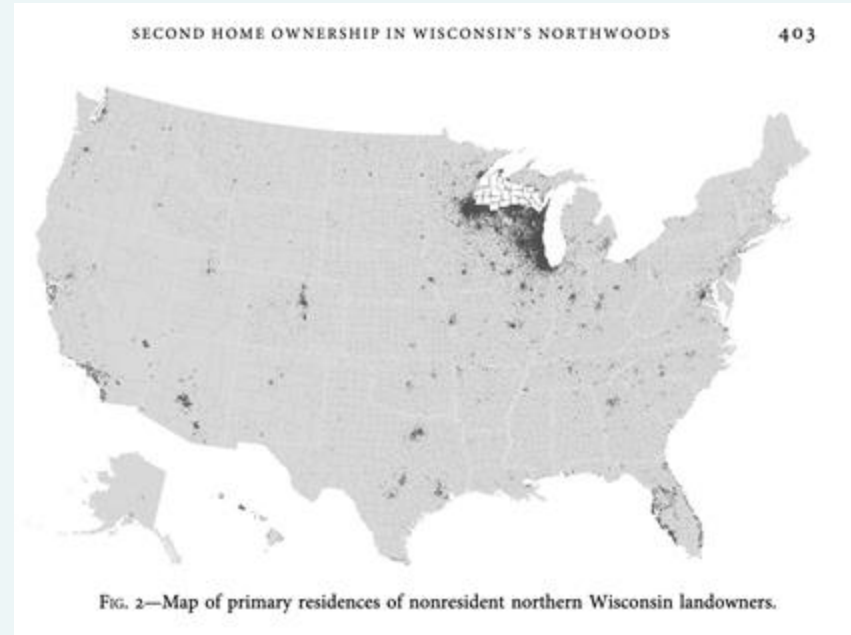
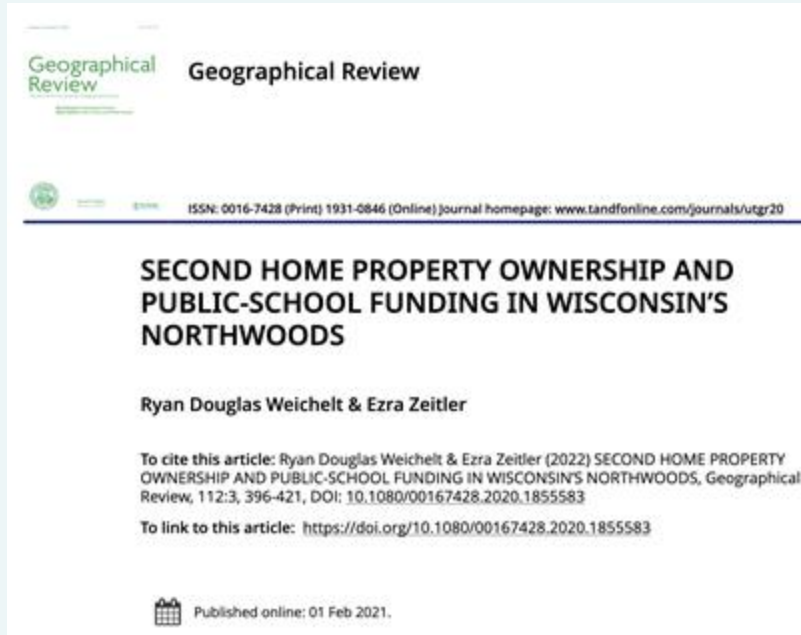


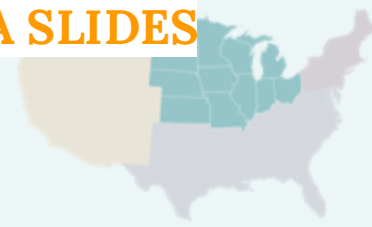


# Spatial Autocorrelation? Not Really



# State Cartographer's Office Parcel Data Study





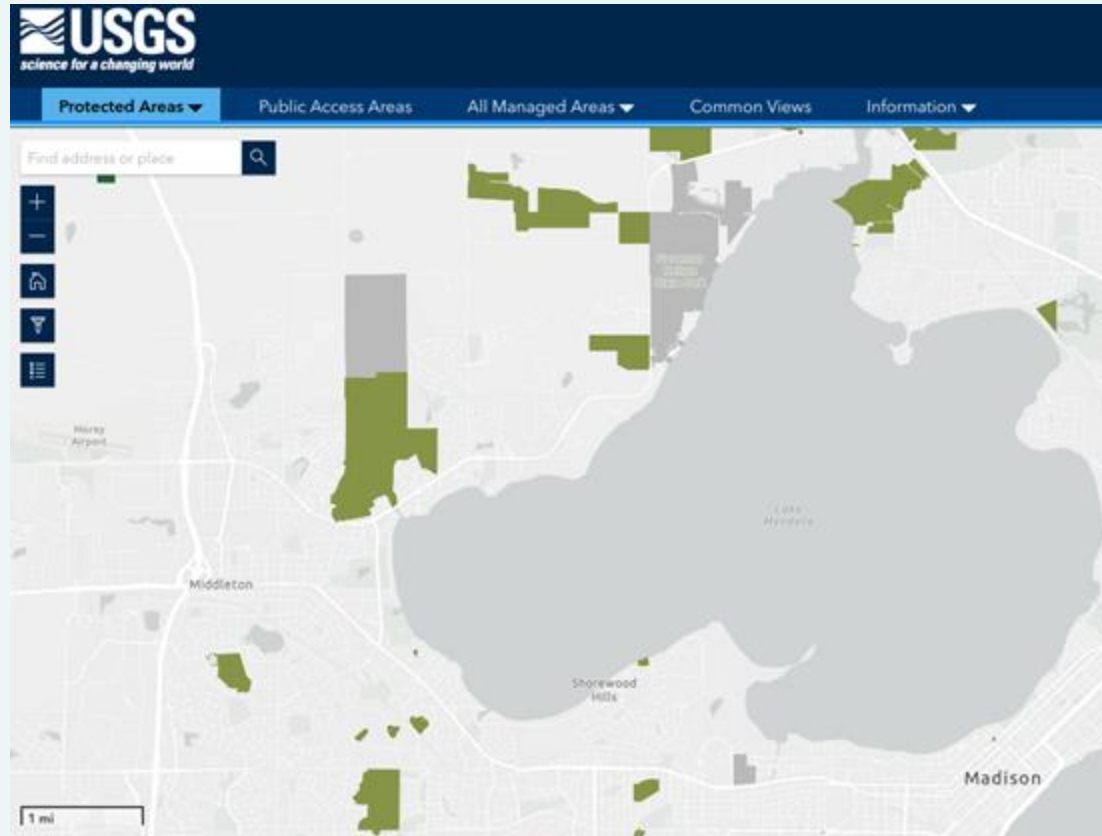
# Reference Links

Corn/field picture <https://archive.alleghenyfront.org/story/corn-ethanol%E2%80%99s-impact-rural-america.html>

Another field picture

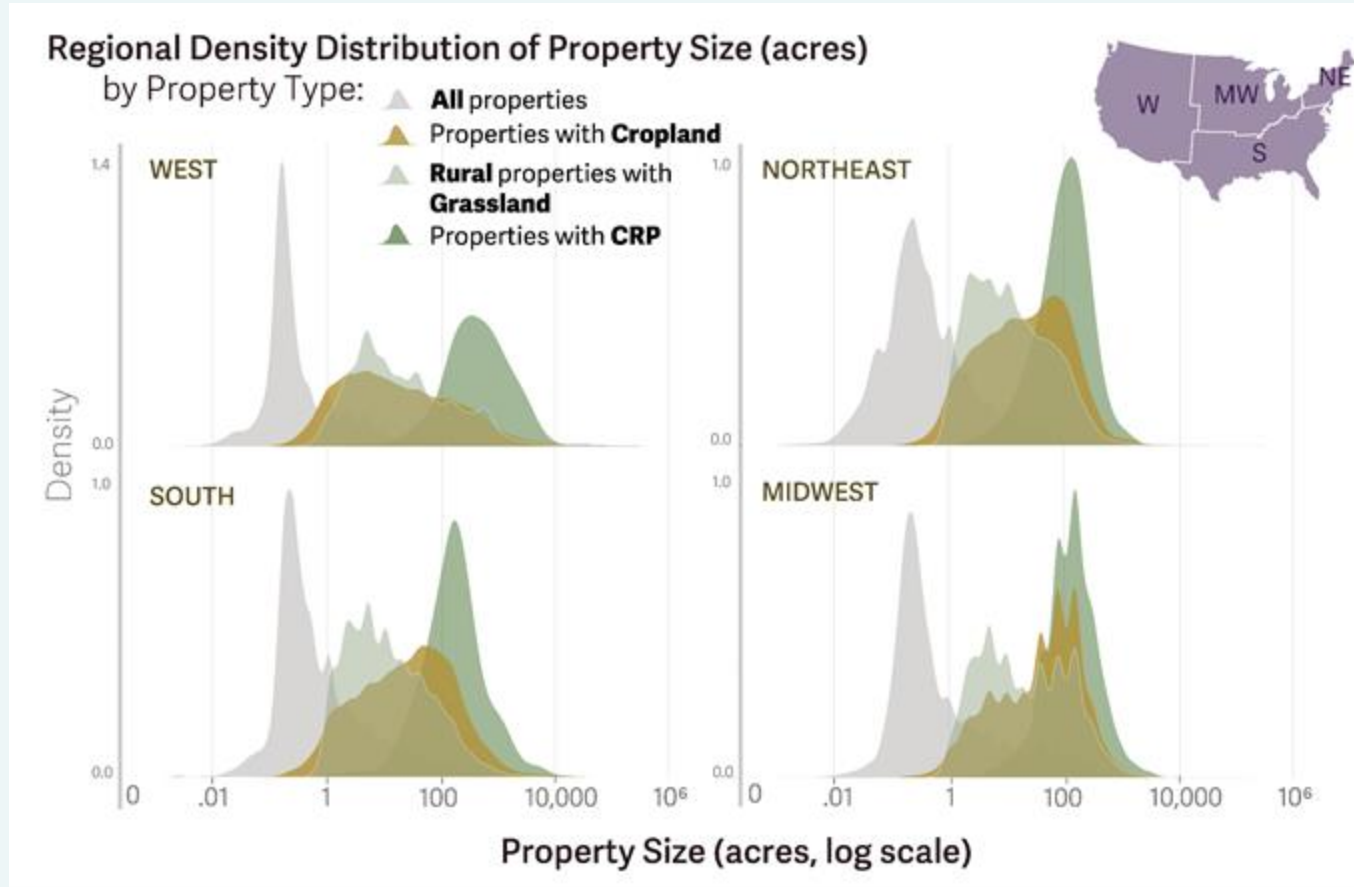
<https://www.ksjbam.com/2022/01/24/u-s-states-with-the-most-organic-farms/>

# Protected Areas in the United States

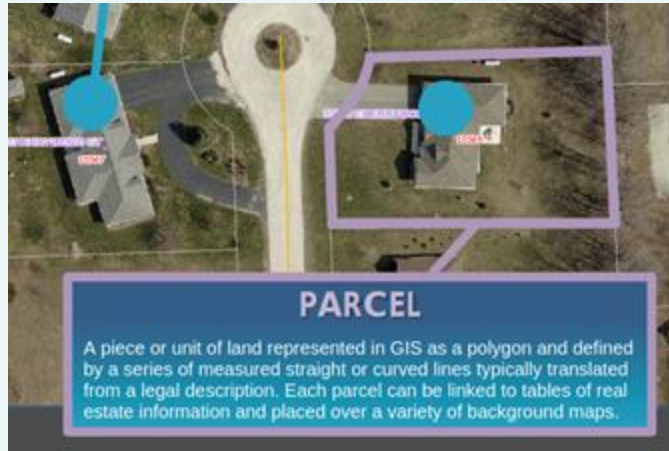




# Example Attribute: Property Size



# (Tax) Parcel Data: Ownership, Mailing Addresses for Taxes



APPMP 2014 Report

Wisconsin Statewide Parcel Map

Search by Address Search by Owner Name Search by Parcel ID

Parcel ID: 070914305010

|                       |                                                     |
|-----------------------|-----------------------------------------------------|
| Parcel ID             | 070914305010                                        |
| Tax Parcel ID         |                                                     |
| Parcel Date           | 02/27/2003                                          |
| Tax Roll Year         | 2024                                                |
| Primary Owner Name    | UNIV OF WIS REGENTS<br>RE DEV & ADMIN UW<br>MADISON |
| Secondary Owner Name  |                                                     |
| Full Mailing Address  | 21 N PARK ST #6101,<br>MADISON WI 53715             |
| Full Physical Address | 970 UNIVERSITY AVE                                  |
| Place Name            | CITY OF MADISON                                     |
| School District       | MADISON<br>METROPOLITAN<br>SCHOOL DISTRICT          |

Map labels: Sewell Social Sciences, Water Science and Engineering Laboratory, Science Hall, North Hall, Ingraham Hall, Birge Hall, Lathrop Hall, Chamberlin Hall, Chadbourne Residence Hall, University Ave, Vilas Hall, Susan B Davis Residence Hall, Teacher Education, Zoology Research Building, Chemistry Building, Wisconsin School of Business, Mergridge Hall, Gordon Strong and Evans Center.

# Why would one own multiple parcels if one is a single land-owning entity?

←  r/Homesteading • 3y ago  
mindheavy

## two adjacent pieces of land, pros/cons to joining them?

Hi everyone, new here, hopefully this is an okay place to ask:

We are about to make an offer on 20 acres. Currently it is two plots with a home on 4 acres and an additional 16 acres that is connected to it but we dont know why it is split.

I am interested to hear some pros and cons to having it separate as opposed to joining the two into a single property.

I have searched around but haven't found much on the topic.

There are no fences or anything separating the two other than the legal description. We're not sure why it was divided this way in the past. Seller is selling it as one property but legally it is two.

 penlowe • 3y ago

They had ag exemption on the 16 but not the 4. Taxing districts can make it worthwhile to do this depending on what your ag is and other strictly-paperwork reasons.

Another consideration to discuss with the tax authority is the ease of dividing and how that plays into your plans. Some places make combining adjacent lots easy, and separating lots hard. Ans vice versa. And some places is no problem at all either way save for zoned lot minimums.

When we purchased our property and were looking into funding the house, one thing that was brought to our attention is that we could "parcel off" say 1-5 acres to put the house on and then leverage the rest of the property as collateral for a loan to build a house on the smaller parcel. Once you pay off the loan, the bank releases their claim on the other acreage. If you

So there's a slim chance that rejoining the 2 parcels may change your zoning designation and thus, your tax rate. But it would be on you to research which zoning would be most beneficial and what you'd have to do to classify. For example, our property was zoned AG when we bought it but the act of buying it flagged the system and converted it to "vacant" until we get it developed at which point it'll go back to AG. So our tax rate went from about \$5 per year to \$200 per year until we get rezoned again and then who knows what a home will do to the estimated value and thus taxes. But the county could probably tell me if I could remember to ask when I'm there. lol

# Creating properties from parcels



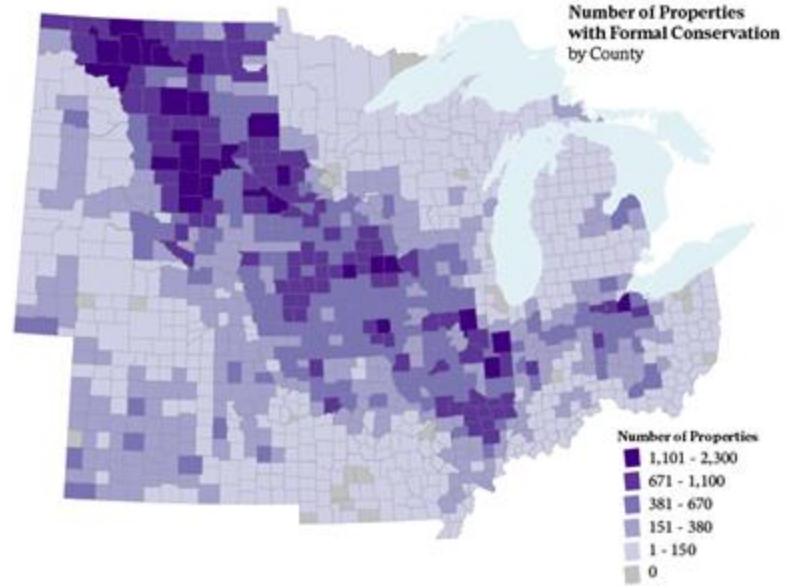
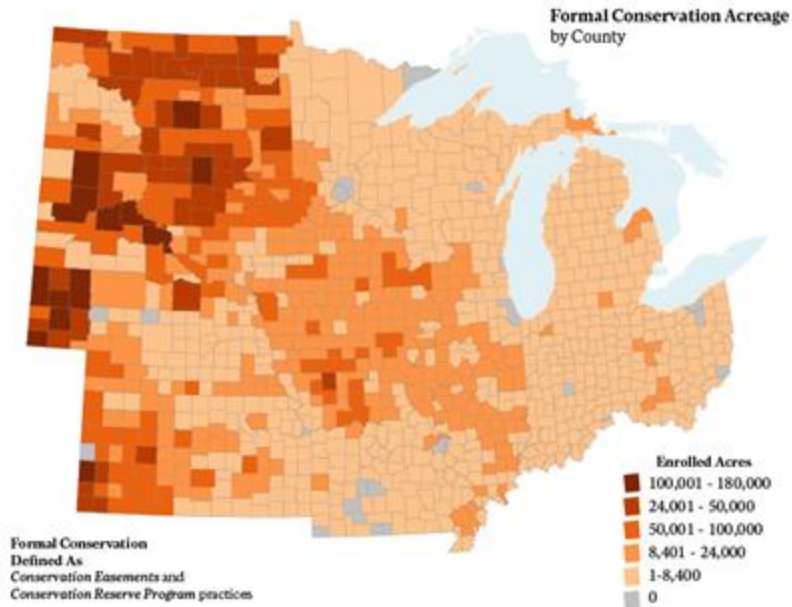
Owner's **Mailing** Address  
(not just parcel street address)



Owner's Name

# Conservation Pathways # 1: Formal

Formal Conservation on Rural, Private Midwestern Properties





# Some highlights:



Grant County sits in the Driftless Area of Mississippi and Wisconsin. This landscape here is because glaciers never reached this area.

Much of the flat land here is used for row crops. In the past, though, steep slopes made it difficult to farm. The meandering stream is a result of the rugged topography.



GRASSLANDS LAND-USE & ENVIRONMENTAL LAB

## Property-Level Characterization of Privately Conserved Lands in the United States

Christina Dennis [clennis@wisc.edu](mailto:clennis@wisc.edu)



GRASSLANDS LAND-USE & ENVIRONMENTAL LAB

## Analyzing Nationwide Parcel- and Property-Level Data for Grasslands Conservation

Christina Dennis [clennis@wisc.edu](mailto:clennis@wisc.edu) June 2025



GRASSLANDS LAND-USE & ENVIRONMENTAL LAB

## Parcels to Properties: Property-Level Analyses for Bioenergy Lands

Christina Dennis, Matt Christie, Tyler Lark, Holly Gibbs



WISCONSIN UNIVERSITY OF WISCONSIN-MADISON

### Background

Understanding the spatial distribution of bioenergy potential is critical for assessing the feasibility of large-scale bioenergy production. This project aims to provide a comprehensive analysis of bioenergy potential across the United States, focusing on property-level data.

### Methods and Data

**Data Layers:**

- Land Use/Land Cover (LULU)
- Soil Type
- Topography
- Climate
- Water Availability

**Processing Flow:**

Raw Data → Data Cleaning → Data Integration → Data Analysis → Results Visualization

### Results

**Summary of Findings:**

- Identified key regions with high bioenergy potential.
- Highlighted areas with high soil fertility and water availability.
- Identified areas with high topographic variability, which may be less suitable for large-scale bioenergy production.

### Preliminary Results

**Summary of Findings:**

- Identified key regions with high bioenergy potential.
- Highlighted areas with high soil fertility and water availability.
- Identified areas with high topographic variability, which may be less suitable for large-scale bioenergy production.

[atlas.glbrc.org](https://atlas.glbrc.org)

See the online atlas, curated by our team to view and download these and other layers.

Integrated Systems Modeling Team